

**PONDMATE: FISH POND MANAGEMENT MOBILE APPLICATION WITH
IOT-BASED AUTOMATIC FISH FEEDER**

A Capstone Project Presented to the Faculty of the
College of Computing and Multimedia Studies
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In Partial Fulfillment of the Requirements
for the Degree of Bachelor of Science
in Information Technology

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ABSTRACT

Title: PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder

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The study PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder aimed to address challenges in semi-intensive fish farming, particularly manual feeding, record-keeping, and monitoring that often lead to feed wastage and suboptimal growth. Using IoT technology and a mobile interface, the system automates feeding schedules, tracks fish growth and pond activities, and calculates operational costs and ROI.

The project adopted the Scrum Agile approach, enabling iterative development of software and hardware with stakeholder feedback. System effectiveness was evaluated using the Technology Acceptance Model (TAM) and the weighted mean approach with thirty (30) fish pond operators, assessing perceived usefulness, ease of use, ease of learning, and user satisfaction. Results showed the system is reliable, efficient, user-friendly, and highly accepted by its users. Across the four constructs of TAM the highest weighted mean score of 4.80 is the satisfaction, while the overall result of weighted mean of the application and device is 4.60.

The study demonstrated that IoT-based fish pond management enhances feeding accuracy, reduces human error, and improves operational efficiency.

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The Researchers

DEDICATION

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Chapter 1

INTRODUCTION

Context of the Study

The rearing, breeding, and harvesting of aquatic and freshwater animals is called aquaculture (Aquaculture, 2025). According to the Food and Agriculture Organization in 2020, as years passed by, the contribution of aquaculture to overall global fisheries is increasing from 25.7% in 2000 to 49% in 2020.

In the Philippines, aquaculture not only supports food security and provides livelihoods but also boosts the national economy by generating valuable export revenue. Coastal aquaculture production is well-established mainly in rural areas (Tahiluddin et al., 2025). Fishes are raised in a controlled environment such as ponds, tanks, or cages in water bodies (Mason, 2023).

Semi-intensive aquaculture denotes production systems that combine natural pond productivity with measured external inputs—specifically, stocking plus at least one of feeding or fertilizing—so that producers augment, but do not fully replace, ecosystem processes to raise yields.

Additionally, the commonly used feeding technique in the Philippines refers to fish farmers manually feeding the fish at a particular time at a certain amount of food (Dada EG et al., 2020). This involves three types - spot, broadcast, and responsive feeding. In spot feeding, feed

is directly deployed on targeted areas of the pond, then all the fish gather and share the feeds.

In this method, growth is optimized while wastage is minimized. On the other hand, the broadcast feeding is a method wherein feeds are thrown and spread in almost all areas of the fishpond.

This method is more suitable for large fishpond operations since it would be difficult for all the fish to gather at one spot. In the responsive method, feed is thrown into the fishpond. Then, fish farmers observe how the fish respond or behave towards the feed. It does not matter whether spot or broadcast feeding is used (Issah, 2024).

Facing numerous challenges that threaten its sustainability and capacity to meet the growing demand for animal protein (Ruben, 2025), the most common problem that fish farmers face is optimizing the feed quantity, as overfeeding is one of the major sources of stress and leads to nutritional imbalance. Proper feeding management is one of the most critical aspects of fish farming. When fish are provided with more feed than they can normally consume and utilize for growth, it can result in several effects, which include decreased feed efficiency, bacterial loads, environmental degradation, and increased mortality rates (Starag, 2023). This was also supported by Issah, who stated that one of the issues being faced by fish farming is the issue of feed. Often, the fish feed is made in

pellets. Overfeeding or underfeeding the fish is a crucial consideration for fish farmers, as underfeeding causes poor fish growth, and overfeeding causes poor water quality (Issah, 2024).

Aside from proper feeding management, nutrient balance also affects the health condition of farmed fish. Excess nutrients not being utilized in fish farms due to a poor feeding strategy are instantly released into the environment, causing secondary pollution (Barbosa et al., 2024).

The study by Nikola et al. (2020) shows that it is possible for aquaculture to increase efficiency by introducing new technologies in the existing process of fish cultivation. According to the study conducted by Francisco et al. (2024), there are existing feeders that effectively dispense fish feed at specific time intervals. In order for the fish farmer to effectively utilize this, they only need to add fish feed to the storage tank, and the feeding process is handled automatically. The College of Fisheries Aquatic Sciences and Technology (CFAST) is facing the same problems as those observed in earlier literature, especially those relating to improper determination of fish feed leading to poor production of the cultured species and manual operations, which could lead to human error. In order to solve these problems and make the process of fish farming more efficient, the

researchers suggest a mobile application that will simplify and optimize the process of managing aquaculture. The system proposed includes the features of scheduling the feeding process, monitoring the costs of operation, and estimating the harvest period. In addition, to further automate the process, the app will integrate with an IoT-controlled automatic fish feeder, where the user can remotely set and manage the time of feeding by means of a mobile interface.

Research Objectives

The main objective of this study is to develop a Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder. Particularly, the study aimed to:

1. determine the existing business process of managing the fish pond operation;
2. define the information requirement in developing the Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder;
3. identify and design the features of the Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder; and
4. evaluate the effectiveness of Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder in terms of:
 - a. Perceived Usefulness;
 - b. Perceived Ease of Use;

- c. Ease of Learning; and
- d. Satisfaction.

Research Questions

The development of the Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder specifically aims to address the following research questions:

1. What is the exiting business process in managing the fish pond?
2. What are the information requirements needed for the development of a Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder?
3. What are the features of the Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder?
4. What is the effectiveness of Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder in terms of:
 - a. Perceived Usefulness;
 - b. Perceived Ease of Use;
 - c. Ease of Learning; and
 - d. Satisfaction?

Scope and Delimitations

The aim of this study is to develop a Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder to optimize and streamline fish farming operations. This research will address the existing system

for managing the client's fish farm by utilizing Business Process Modeling (BPM). It will also determine the features and information requirements needed for the proposed system by utilizing Information Requirements Analysis (IRA). Finally, the effectiveness of the application, integrated with IoT technology, will be evaluated by applying the Technology Acceptance Model (TAM).

The scope of the study includes identifying and analyzing the operational workflows of CFAST Aquaculture Farm, located in Barangay San Roque Mercedes, Camarines Norte, which currently employs manual preparation, management, and harvesting of fish.

Additionally, the study involves requirements analysis that revolves around: data on the quantity of feed per feeding routine, frequency of fish feeding, a report of expenses from the preparation of the fish pond until harvesting, and production activities.

The application will be available through online access and has two types of users: fish pond caretakers and owners/managers. The functionality of the IoT device is limited to the following features: scheduled and on-demand feeding, and monitoring of feed levels.

Alongside efficient monitoring of fish feeding, the application will also offer return on investment (ROI) analysis, production cost analysis, and an estimated

harvest time for the fish. The evaluation of system effectiveness will be controlled in a testing field during the study period, from January 2025 to December 2025, and will be based on the Technology Acceptance Model.

However, this study has certain limitations. The proposed project is designed only for small-to-medium-scale fish farms. Although the application will have options for selecting different breed species, this will be restricted to two aquatic types specifically: bangus (milkfish) and tilapia. In the event of a natural disaster, the clients stated that fish pond operation shuts down, conducts emergency harvest, and production is discontinued until they fully recover, which is beyond the researchers' objective of automating the fish feeding. Hence, the feature offered by the proponents is to generate pond history, showcasing the events and archive of ponds. It also provides an option to stop pond production if a disaster happens or if the owner desires to, for any reason.

External factors such as power outages, internet reliability in rural areas, and regulatory compliance with fish feeding devices and placement will not be delved into in this study. The provided scope and delimitation of the research prove that this will be focused and ensures the feasibility of success within the stated timeframe, applicability of findings to stakeholders.

Significance of the Study

This mobile application can significantly benefit the stakeholders involved in aquaculture:

College of Fisheries Aquatic Sciences and Technology.

As the project's main client, CFAST will benefit from an automated and more efficient feeding system. The app helps them reduce manual work, avoid overfeeding, and improve cost and profit monitoring.

Fish Pond Owners/Manager and Caretakers. The system offers a user-friendly way to manage feeding schedules, monitor pond activities, and track expenses all from a mobile device.

Aquaculture Industry. By promoting modern tools such as this app, the project supports the modernization of fish farming. It contributes to more sustainable and tech-driven aquaculture practices.

Future Researchers. This study can serve as a reference for future developers and researchers who aim to combine mobile technology and IoT in solving real-world problems in agriculture and aquaculture.

The Researchers. Through this project development, it strengthens both academic knowledge and practical skills in Information Technology for the whole group. It also promotes collaboration and real-world application of theoretical concepts among the members.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of related literature collected from various sources that are relevant to this study. It includes existing publications, previous research, and studies by different experts and individuals that support and relate to the core focus of the project.

Existing Fish Pond Management Systems

Diving into the current existing fish pond business process, in a journal written by the Southeast Asian Fisheries Development Center Aquaculture Department, the pond culture of tilapia was discussed. There are four stages in the production; first is the pond preparation, which is done through drying, leveling, and, if necessary, liming the bottom. Moreover, the elimination of fish predators and competitors are handled together with the growing of natural food, such as lab-lab and plankton. For lab-lab, chicken manure (1-2 tons/ha) and diammonium phosphate (18-46-0, at 1-2 tons/ha) are used as fertilizers, while in the plankton method, the pond is fertilized with 50 kg/ha of ammonium phosphate (16-20-0) every two weeks.

Following this, when the growing of natural food (lab-lab and plankton) is available about 1-2 weeks after fertilization, the next stage begins with the stocking of

fingerlings. Afterward, during pond management, adequate food and maintenance of good water quality are essential, along with fertilization or feeding applied as recommended. Finally, in the stage of harvest and marketing, depending on the size desired for the market—which ranges around 100–200 g per fish—fish are harvested after three or four months.

Moving on to the survival aspect, according to the Philippine Tilapia Industry Roadmap by the Bureau of Fisheries and Aquatic Resources (2023), the survival rate in semi-intensive culture systems typically ranges from 50% to 80%. Likewise, the Philippine Milkfish Industry Roadmap 2021–2040 indicates that milkfish grow-out systems can achieve survival rates of around 90%, particularly under semi-intensive culture conditions (Bureau of Fisheries and Aquatic Resources, 2021). These data highlight the importance of culture system management in ensuring productivity and profitability in aquaculture operations.

In addition, the Tilapia Production Guide by the Bureau of Fisheries and Aquatic Resources (BFAR) (2020) provides comprehensive guidelines for tilapia farming in the Philippines, emphasizing semi-intensive culture systems. The guide details essential production practices, including site selection, pond preparation, stocking densities, water quality management, feeding protocols, and harvesting techniques. In semi-intensive systems, tilapia

is stocked at densities ranging from 4 to 8 fish per square meter (equivalent to 40,000 to 80,000 fish per hectare), with supplemental feeding commencing in the second month and continuing until harvest. The fertilizer is also calculated, as fertilizer use in semi-intensive pond aquaculture in Southern Telangana accounted for only 0.48% of the total variable costs, showing minimal reliance on chemical inputs compared to major expenses such as feed and fingerlings (Sindhu et al., 2023). Latino et al. (2020) also applied Business Process Model and Notation (BPMN 2.0) to improve efficiency and sustainability in aquaculture operations. Their study, "Business Process Modeling in Aquaculture for Environmental Sustainability and Fish Traceability," demonstrated how BPMN can map and optimize fish production processes, integrate IoT-based monitoring, and reduce manual work through digital process management.

In conclusion, the fish pond production process for tilapia and milkfish involves a systematic yet labor-intensive set of stages—from pond preparation to harvest—requiring careful management and consistent monitoring. The referenced studies and roadmaps by SEAFDEC and BFAR, alongside the 2020 BPMN-based aquaculture study by Latino et al., collectively emphasize how process modeling and digital management tools can enhance productivity, sustainability, and traceability in modern aquaculture operations, ensuring long-term viability and scalability.

Information Requirements for Fish Pond Management System

In recent years, the integration of automation and information systems has significantly enhanced fish pond management efficiency. Hossain et al. (2021) developed a fish-farming activity calendar that outlines essential tasks such as pond preparation, fingerling stocking, grow-out management, and harvesting. This advancement demonstrates the increasing shift from traditional manual processes toward more structured and data-driven aquaculture management.

Given the critical role of feeding schedules in fish growth and productivity, the need for automated solutions in pond management has become increasingly evident. For instance, Mahendra and Sunardi (2023) developed an IoT-based feeding system for catfish farms, enabling remote monitoring of feed levels and automated feeding schedules through the Blynk application. This innovation reduced manual labor and improved feeding efficiency. Building upon this concept, Faiz et al. (2022) introduced an IoT-based solar-powered automatic fish feeding system that allowed farmers to manage feeding operations remotely through a mobile app and dashboard. Such digital platforms offer key monitoring features, including real-time sensor data, fish production cycle tracking, daily feeding logs, and fish health monitoring—enhancing data-driven decision-making in aquaculture. Similarly, Kiranmayi et al. (2021) developed

an Android-based mobile application that provides fish farmers with centralized management information, focusing on optimal feeding strategies and implementing best practices. Complementing this, Adimulam et al. (2024) introduced an Automatic Feed Dispenser for Aquaculture utilizing Arduino technology, featuring both automatic and manual modes. Integrated with ultrasonic sensors and Bluetooth modules, this system minimizes human intervention while improving feeding accuracy and operational efficiency. Furthermore, Divina et al. (2023) advanced the field through the development of the FISDA Smart IoT-Based Fish Farming Monitoring System, which combines real-time water quality monitoring and automated feeding into a unified platform—enhancing productivity and profitability. Likewise, an aquaponic monitoring system with an automatic fish feeder built using Arduino and Wi-Fi technologies achieved an average performance efficiency of 92.5%, demonstrating its reliability in real-time pond management and remote notification systems.

At the international level, advanced technologies such as those developed by Observe Technologies and the Umitron Cell system in Singapore and Japan leverage artificial intelligence (AI) and remote control capabilities to optimize feeding schedules, reduce feed waste by approximately 21%, and maintain water quality—contributing to more sustainable aquaculture practices.

The integration of automation and machine learning in fish farming has been shown to improve profitability by lowering daily operating costs and enhancing operational efficiency. Since feeding costs represent the largest portion of production expenses, effective feeding strategies that optimize feed distribution not only promote fish growth but also mitigate environmental impacts caused by feed wastage (Chrispin et al., 2020). Traditionally, manual feeding methods have been inefficient due to uneven distribution; however, the use of Automatic Fish Feeders (AFF) has led to measurable improvements. Studies by Susilawati et al. (2023) show that AFF implementation can lower the Feed Conversion Ratio (FCR) from 1.44 to 1.15 and increase Feed Efficiency (FE) from 69.4% to 86.8%. These findings emphasize the role of automated feeding systems in reducing production costs while enhancing sustainability and profitability.

To add, Paguia et al. (2020) found that vermicomposting in an integrated farming system in Bataan yielded a 30% ROI, proving its profitability. In semi-intensive aquaculture systems, stocking density plays a critical role in optimizing fish growth and maintaining water quality. According to the Food and Agriculture Organization (FAO), the recommended stocking density for milkfish (*Chanos chanos*) ranges from 0.8 to 1.2 fish per square meter, while for tilapia (*Oreochromis spp.*), the

suitable range is 2 to 4 fish per square meter. These stocking rates are designed to balance production efficiency with environmental sustainability, allowing sufficient space for growth and minimizing stress and competition among fish (FAO, 2021).

Ultimately, the growing body of research from 2020 to 2024 demonstrates a clear trend toward digital transformation in aquaculture. These advancements collectively illustrate how technology-driven systems can reduce manual labor, improve feed utilization, enhance monitoring accuracy, and ultimately contribute to the long-term profitability and environmental sustainability of aquaculture operations.

Features of Fish Pond Management System

In fish farming, feeding is an essential aspect that affects the entire production process. With IoT-based automated feeding systems, farmers can dispense the exact amounts of pellets or feeds at the desired time and schedule according to the fish's needs (Mohamed et al., 2025). Timed feeding controls promote sustainable fish growth and efficiency by mitigating overfeeding, while commercial pellet feeds provide necessary nutrients but can also negatively affect pH levels (Chen et al., 2022). This makes the system an automatic or remotely controlled feeding mechanism that ensures accurate scheduling and minimizes waste (Rathy & Jenefer, 2024). Moreover, water quality

monitoring, automatic feeders, feeding control systems, and smart marketplace platforms are just a few of the IoT technologies used in aquaculture management that significantly improve the process (Yusoff & Hanafi, 2025). Several studies have demonstrated the effectiveness of fish feeding systems, highlighting their efficiency and operational convenience. These systems allow real-time monitoring of feeding activities, enabling timely interventions and customization of feeding schedules based on factors like feeder size (Son & Jeong, 2024).

Further advancements have led to the development of efficient semi-automatic systems designed to promote the healthy growth of aquatic organisms. Such systems include feeding mechanisms and monitoring of water quality parameters using sensors for pH and temperature, which automatically dispense feed according to the fish's requirements (Indrawati et al., 2025). The power of IoT continues to enhance the efficiency of maintaining aquariums and fish ponds by automating critical tasks and transmitting real-time data to a user's smartphone (Jibon et al., 2024). In aquaponics, IoT technologies enable real-time data collection and automation of interventions for optimal plant and fish growth in prototype systems (Tandel, 2025). IoT devices equipped with advanced sensors and internet connectivity improve traditional feeding methods by ensuring precise and timely feeding schedules, promoting

sustainable practices, and improving pond quality (Yusoff & Hanafi, 2025).

Intelligent aquaculture further utilizes advanced equipment and robotics to perform breeding, water treatment, feeding, monitoring, net washing, and animal classification—enhancing production efficiency, product quality, and safety while reducing costs (Tandel, 2025). However, effective fish feeding remains complex, as it involves parameters such as fish type, density, water quality, and temperature, which, if unmanaged, can lead to fish mortality or poor health (Indrawati et al., 2025).

To address these challenges, proposed IoT-based systems use ESP32 sensors to collect data on fish health, monitor activity, and refill water using servo motors, thereby reducing feeding inefficiencies and mortality rates (Mohamed et al., 2025). In the development of aquaculture, cost and revenue play critical roles in determining profitability—high costs due to the culture system or location can lead to uneconomic conditions, while strong revenue performance enables farm expansion and sustainability (Jibon et al., 2024). Effective financial management is essential, as all financial activities must be documented. Feeding patterns, fish growth, and feed consumption can be easily monitored by farmers using IoT systems, making record keeping crucial for tracking progress, optimizing operations, and identifying issues

early (Rathy & Jenefer, 2024). For managing daily operations, calculation of the daily feed quantity per tank is necessary by multiplying the biomass of the tank by the coefficient from its feeding table (Indrawati et al., 2025).

Apart from this, the number of hours the species should be fed and how many times in a day should be taken into consideration. This helps in generating the dosage and sending it to each tank. Fish farms need technology for control and monitoring, the study conducted by Al-mutairi et al. (2023) achieves the best performance in fish pond in real-time monitoring and control of systems through the use of IoT based systems. To control fish growth and increase productivity, the ponds contain a node connected to sensors, actuators, and wireless communication modules. Also, controllers are set for water and environment quality.

Accurate measurement of fish feed intake is critical in measuring accuracy. The proposed system by Adegboye et. al (2020) shows that dispensing feeds without human intervention is successful through the use of fish behavioral vibration analysis and artificial neural works. Similarly, Rizal et al. (2025) uses IoT based technology accompanied by Blynk platform that ought to solve the decrease in quality and quantity of fish that is brought by undisciplined feeding. The result of the study shows

that accurate fish feeding is provided by the infrared distance that total to 5,1% error, while the success of the infrared sensor is visible on throwing feed with 100% success rate.

Chiu et al. (2022) proposed a system with different IoT devices to enable real-time data collection, the model resulted in a successful autonomous feeding system eliminating a high percentage in leftover feeds. This helps aquaculture operators in assisting and managing different fishpond remotely.

In Taiwan climate and lack of human resources causes huge aquaculture losses, hence Chen et. al (2022) designed an IoT device transmitting data wirelessly with various sensors for temperature, pH level, dissolved oxygen, water level and fish's life expectancy.

The system can work 24 hr per day reducing losses and data errors. Building on these innovations, the integrated system made by Ravi et al. (2023) is an IoT-based fish pond water monitoring system that consists of real-time sensors that track and send inputs to a micro-controller to be stored in a real-time database, through an application users can monitor these values effectively and quickly whenever needed. Although it is transformative, IoT also comes with limitations that need areas of advancement. Such as corrosion of sensors, limits in data fusion, and impacts of environmental transmission. Finally, Yusoff

(2025) developed a study using IoT-based device and an android-based mobile application, parameters of aquaculture's basic needs and environment of fishes will be monitored, this device consist sensors that detects level of pH, water temperature, dissolved oxygen and ammonia level to know the requirement for proper fish farming.

Overall, the growing integration of IoT, automation, and intelligent systems in aquaculture marks a significant step toward modern, data-driven fish farming. These technologies enhance feeding precision, optimize water quality, and enable real-time monitoring, which collectively improve productivity, sustainability, and profitability.

Evaluation of Fish Pond Management System

The Technology Acceptance Model (TAM) is used in the article by Kharina, et al. (2023) to examine people's attitude toward adopting financial technology shown in Figure 1. Perceived usefulness and perceived ease of use are the highlights of the study which significantly influence user's attitude and intentions. Several studies explored innovations and system evaluations in aquaculture and fish feeding technologies. Ouko et al. (2022) investigated the use of black soldier fly larvae meal as an alternative fish feed, illustrating how the Technology Acceptance Model (TAM) applies to both non-ICT aquaculture

innovations and digital/IoT tools. Leong et al. developed and tested an automated fish counting system using Convolutional Neural Networks (CNN) and Contrast Limited Histogram Equalization. They evaluated its performance through usability testing, performance metrics (accuracy, recall, precision, and F1-score), and comparative analysis.

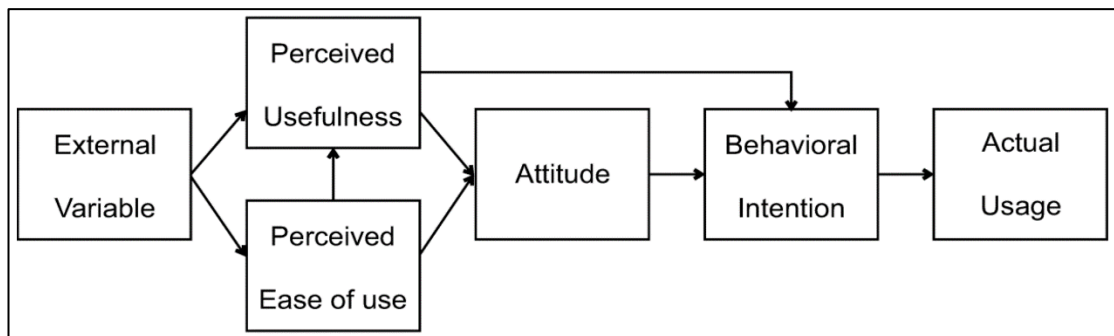


Figure 1. Technology Acceptance Model

Similarly, another study on a pH monitoring and automated clownfish feeding system based on IoT and fuzzy logic achieved an 87% user acceptance rate in usability testing. Khater et al. (2021) evaluated a locally manufactured fish feeder under various conditions, analyzing feed productivity, pellet size, air flow, and screw speed. Efficiency improved up to 540 rpm, with reduced energy consumption and operational costs ranging from 0.09 to 0.16 EGP per kg of feed. Chen et al. (2022) assessed an automated fish-feeding system using CNN and GRU neural networks, applying metrics such as accuracy, precision, recall, and F1-score, while also evaluating energy efficiency. Kumar et al. introduced the Fish Feeding Automation and Aquaponic Monitoring System (FFAAMS),

integrating real-time monitoring, data-driven decision-making, and IoT-based remote control for optimal feeding and environmental management. Finally, Fasoyin et al. (2021) evaluated a small-scale mechanical fish feeder using five tools: experimental testing, dispensing efficiency (93.1% at 50 rpm for 2 mm feed), throughput capacity (75.76 kg/h at 80 rpm), ANOVA statistical analysis, and cost evaluation, confirming its affordability and suitability for small-scale use.

Drawing from the reviewed studies, the Technology Acceptance Model (TAM) demonstrates its relevance not only in financial technology but also in aquaculture innovations. Researchers applied TAM to assess user attitudes, perceived usefulness, and ease of use in adopting automated feeding systems and IoT-based tools for fish farming. Various technologies—such as CNN-based fish counters, IoT-enabled feeders, and fuzzy logic systems—showed high efficiency, precision, and user acceptance. These findings emphasize that technological adoption in aquaculture relies heavily on both usability and performance, highlighting TAM's effectiveness for evaluating innovation acceptance in this field.

Synthesis of the Art

This part introduces the similarities and differences between the available literature and the current study, which is the development of fish pond management, using

traditional methods to an IoT-based automation and digital transformation. The review brings out the way past studies have handled the issues of production efficiency, feeding management, monitoring and evaluation- besides pointing out the gaps that exist which should be filled by the proposed project.

According to the Southeast Asian Fisheries Development Center (SEAFDEC) and Bureau of Fisheries and Aquatic Resources (2020-2023), fish pond production is a labor-intensive process that consists of several stages: preparation of ponds, their stocking, management, and harvesting. Through these studies, the issue of appropriate feeding, water quality and environmental control were used to realize a survival rate of 50%-90% under semi-intensive systems. However, even these traditional approaches are still reliant on manual processes and human error, and they may lead to inefficiency and unevenness.

Contrarily, the Business Process Modeling and Notation (BPMN 2.0) proposed by Latino et al. (2020) was used to realize the map of the aquaculture processes and optimize them to decrease the number of individuals and enhance the effect on the environment. Where SEAFDEC and BFAR paid attention to the biological and procedural parts of the pond management, Latino et al. considered the aspect of process optimization and digitization. It represents a change of traditional to data-based aquaculture but does

not have a direct application in the localized small scale pond management systems.

Further research revealed the increasing use of internet of things and automation to address the historical problems. An example is the study by Hossain et al. (2021), who suggested using a fish-farming activity calendar to organize the farming process, and Mahendra and Sunardi (2023) and Faiz et al. (2022), who created an IoT-based feeding system allowing to control feeding remotely, monitor all processes, and use less energy in the form of mobile apps. Likewise, Kiranmayi et al. (2021), Adimulam et al. (2024) and Divina et al. (2023) developed Android- and Arduino-based systems centralizing pond controls, feed automation, and measuring environmental parameters- representing one step towards smarter and more sustainable aquaculture.

These systems are varied in the way they are implemented in technology, but they are all geared towards the same objective which is to reduce labor, maximize feeding, and maximize productivity. These efforts were then further advanced with international innovations such as Observe Technologies and Umitron Cell in Singapore and Japan where AI was used to optimize feeds, minimized waste by about 21, and ensured water quality (Chrispin et al., 2020). Nevertheless, the prevalence of such studies revolved around large-scale and industrial aquaculture

farming as opposed to small- medium local farms.

However, the automation and IoT functionality was not incorporated into the design of the study that created a gap in the centralization of process modeling, real-time monitoring, and automated feeding in a single platform.

In the case of system evaluation, performance testing, usability evaluation, and cost analysis were the methods used to evaluate automated feeding systems by Khater et al. (2021), Leong et al. (2024), and Fasoyin et al. (2021). In the meantime, Kharina et al. (2023) used the Technology Acceptance Model (TAM) to evaluate perceived usefulness, ease of use, and user attitude-another method which has also been successfully implemented to evaluate aquaculture technologies (Ouko et al., 2022).

Overall, the available literature demonstrates that, although automation, IoT, and BPMN have made fish farming more efficient and sustainable, they are still disjointed, with some of them only considering feeding automation, monitoring, or business process modelling. The proposed study will fill these gaps by creating a comprehensive Fish Pond Management Mobile Application with IoT-based automatic feeding, real-time data monitoring, and the analytical features of IRA and test them within the framework of the Technology Acceptance Model (TAM).

Chapter 3

METHODOLOGY

This section outlines the procedures and systematic approach used to conduct this study. Detailing the software development methodology, schedule of activities, sources of data, population of the study, data collection method, data analysis, statistical treatment of data, and ethical consideration. By clearly describing these steps, the methodology of this project ensures the findings are valid, can be confidently interpreted and provides transparency.

Software Development Methodology

In case of PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder, Scrum Agile will be applied as the software development approach to facilitate flexibilities and iterative development, which is an Agile methodology shown in Figure 2.

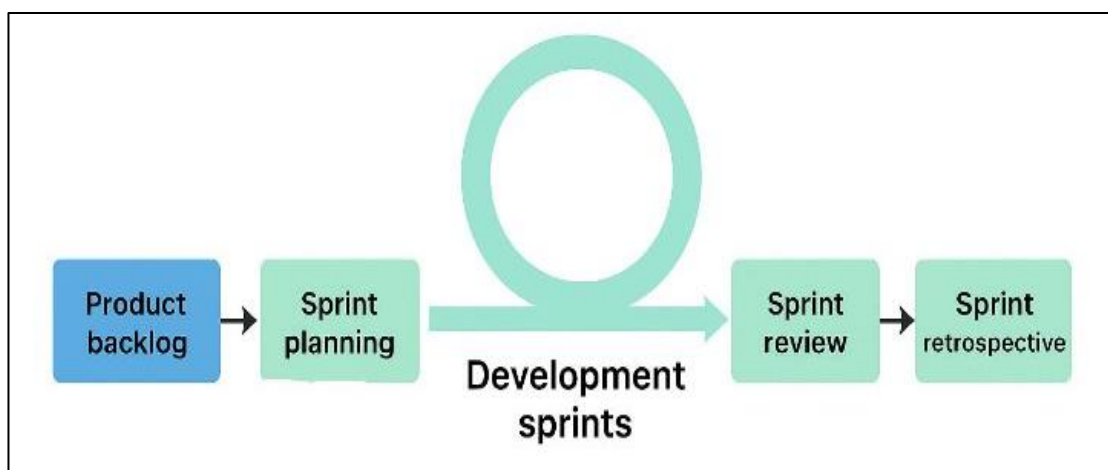


Figure 2. Scrum Agile Software Development

Rachmawati et al. (2023) proved that the application of Scrum with a well-organized sprint cycle improves

teamwork, speeds up the development process, and aligns user-driven improvements across the project life cycle.

Firstly, the product backlog is a prioritized list of all system requirements and tasks. In this project, requirements were collected through stakeholder interviews with fish farmers and pond managers and transformed into user stories, including automated feeding schedules and pond monitoring, which served as the basis for sprint planning. Next, sprint planning is the phase where the team selects backlog items for the upcoming sprint and defines goals and deliverables. For this project, tasks were scheduled starting with UI wireframes, system design, and database setup, followed by automated feeding and IoT integration. Then, a development sprint is where the team implements planned tasks and builds working increments of the system. Early sprints focused on software setup, while later sprints included hardware setup and software functionality. Afterward, the sprint review is when completed work is presented to stakeholders for feedback, which in this project included demonstrations of software and hardware functionalities. Lastly, the sprint retrospective is where the team reflects on successes and areas for improvement, focusing on processes, tools, and collaboration rather than just mistakes. In this project, the team documented what worked well, such as successful hardware integration and stable IoT communication, and

identified areas for improvement, such as optimizing power distribution, refining code.

Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder Schematic Diagram Explanation

Figure 3 presents the complete system architecture and wiring layout of the PondMate: Fish Pond Management Mobile Application with IoT-based Automatic Fish Feeder. It provides a clear overview of how the hardware and software components are interconnected to each other.

The development of the system involved configuring the Arduino Mega, which acts as the central controller, managing all connected devices and modules. The ESP8266 Wi-Fi module enabled the device to communicate with the mobile application, transmitting real-time pond data and receiving feeding commands remotely.

The load cell with HX711 module monitors feed quantity, allowing accurate measurement. Sensor readings are processed by the Arduino Mega and simultaneously sent to the cloud server via the ESP8266, enabling remote monitoring and data logging on the mobile application.

The LCD display provides local feedback, showing feeding schedules and pond data. Power is supplied through two 12V batteries, regulated via fuses and step-down modules to ensure safe operation of all components.

Overall, the PondMate system integrates IoT-enabled hardware with a mobile application, enabling automated

feeding, real-time pond monitoring, and remote management, reducing manual labor, improving feeding accuracy, and enhancing fish pond productivity.

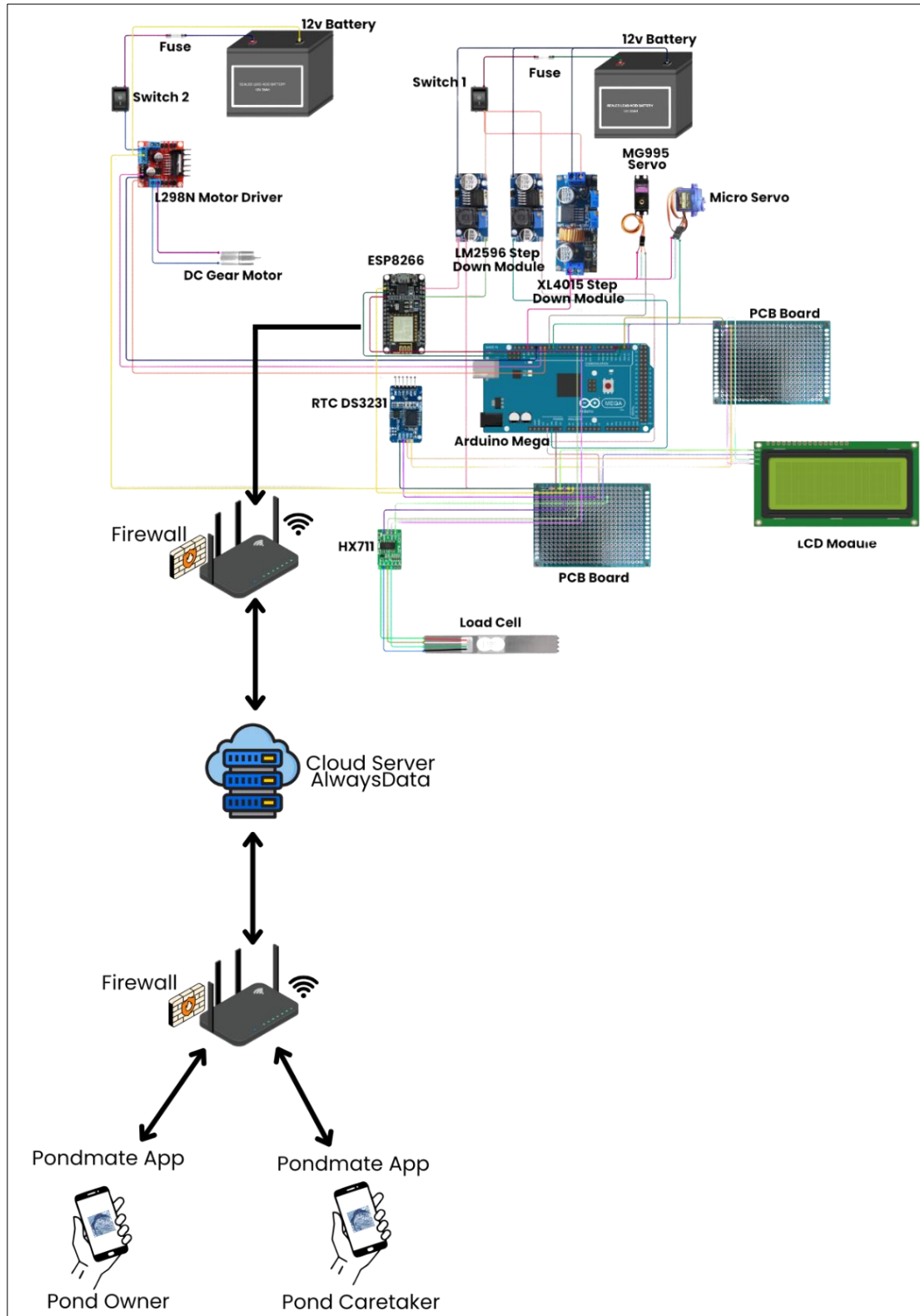


Figure 3. Schematic Diagram

Schedule of Activities

The following schedule of activities in Table 1 outlines the tasks and corresponding timeline for the development of this project within the time span of February to December, based on the 12 three-and-a-half-week sprints of the Scrum Agile framework.

A clear timeline spanning from February to December is structured from the table given below, divided into monthly periods and focusing on sprints to ensure a thorough development and steady progress of the project. The initial phase begins with stakeholder engagement and project planning in February establishing objectives and aligning expectations. In this phase, the project team will focus on collecting detailed requirements and understanding the needs of all stakeholders. Structured interviews, observations of current fish farm operations, and workflow documentation will be conducted to identify both functional and non-functional requirements. This process ensures that the system will address real operational needs and align with stakeholder expectations. Additionally, the team will use the gathered information to refine the project scope, prioritize features, and prepare a comprehensive product backlog for subsequent development sprints. This phase lays the foundation for designing a system that is practical, effective, and aligned with the goals of the fish farm operations.

In Sprint 1 , the researchers will identify and engage with key stakeholders, conducting initial coordination meetings. The project scope, objectives, and success criteria will be defined, while a high-level project roadmap and communication plan will be prepared to guide subsequent development activities.

Subsequently, in Sprint 2, the researchers will conduct comprehensive requirements gathering to ensure that the system addresses actual operational needs. Structured stakeholder interviews, observations of current fish farm practices, and documentation of user workflows will be undertaken to identify functional and non-functional requirements.

Following the initial requirements gathering, Sprint 3 will focus on requirements validation and technical evaluation. The researchers will review and finalize the collected requirements with stakeholders, resolving any ambiguities or inconsistencies. Features will be organized and prioritized into a product backlog to establish a clear development roadmap. Simultaneously, a technical spike will be conducted to evaluate and select appropriate technologies for mobile application development, backend infrastructure, database management, and IoT integration.

In Sprint 4, the researchers will translate the validated requirements into comprehensive system designs. User interface wireframes will be developed to define the

layout and navigation flow of the application, while Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD) will be created to model system processes and database structures. By the end of this phase, the researchers will have a detailed design blueprint, which will serve as a guide for development and ensure consistency with stakeholder expectations.

During Sprint 5, the researchers will acquire and verify the required IoT components and configure the development environment to support coding and testing. Backend authentication mechanisms and an online MySQL database will be implemented to establish a secure and reliable system infrastructure. Completing this setup phase will ensure that subsequent development activities can proceed efficiently and without technical obstacles.

In Sprint 6 , the researchers will develop core system functionalities. This will include creating the feeding schedule interface, programming feeding logic, and integrating calendar-based activity tracking. These foundational features will enable effective management and monitoring of feeding operations.

Following this, in Sprint 7, the researchers will integrate the mobile application with the IoT-based fish feeder hardware. Real-time communication protocols will be configured, and reliability and performance tests will be conducted to ensure stable and accurate operation. This

phase is essential for validating the practical functionality of the system in real-world operational conditions.

During Sprint 8, the system's functionalities will be expanded to include production cost tracking, expense management modules, and ROI computation. These features will provide users with tools to monitor operational costs and evaluate financial performance, enhancing the system's utility and relevance to fish farm management.

In Sprint 9, the researchers will implement monitoring and record-keeping features, such as harvest estimation and feeding logs, enabling users to review historical feeding activities.

Subsequently, in Sprint, refinement and security enhancements will be applied. The researchers will improve the user interface and overall usability, strengthen access control and data protection measures, and resolve software defects.

In Sprint 11, the researchers will conduct comprehensive system testing, including functional, integration, and usability testing, to verify that all components meet the defined requirements. Final user manuals, technical documentation, and maintenance guidelines will be prepared to support effective system usage and long-term sustainability.

Finally, in Sprint 12, the system will be deployed to

the final test environment during the third week of November, and operational readiness will be verified. In the last week of November, the researchers will conduct a system demonstration for stakeholders and perform a formal evaluation to ensure that all project objectives have been met. From the first to last week of December, the approval process will be completed, the final defense conducted, and formal project sign-off secured, marking the official completion of the study. This final phase ensures that the system is validated, approved, and ready for operational implementation, demonstrating the successful achievement of the research goals.

Table 1

Schedule of Activities

ACTIVITIES	February	March	April	May	June	July	August	September	October	November	December
Project Planning and Initial Stakeholder Engagement	■										
Requirements Gathering and Stakeholder Interviews	■	■									
Sprint 1: Requirement Finalization and Planning		■	■								
Sprint 2: UI Wireframes, System Design, and IoT Acquirement			■	■							
Sprint 3: User Authentication and Online MySQL Setup				■	■						
Sprint 4: Feeding Schedule Development + Logic					■	■					
Sprint 5: IoT Integration Components Setup						■	■				
Sprint 6: Feeding Control Logic and Real Time Commands							■	■			
Sprint 7: Production Cost Tracking and ROI Computation								■	■		
Sprint 8: Harvest Estimation Feature									■	■	
Sprint 9: Notifications, Feeding Logs, and History									■	■	
Sprint 10: Security Improvements and Internal Testing										■	■
Sprint 11: User Feedback											■
Sprint 12: Final Testing, Documentation, and Refinement											■
Deployment: Final Test Sites, Presentation, and Closure											■

Sources of Data

In order to fulfill the above-mentioned goals, the authors of PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder collected both primary and secondary data to inform the analysis of the system, its design, and evaluation.

To reach Objective 1, which aims at establishing the current business process of fish pond operations management, primary data were collected using interviews and direct observation that will be done at the College of Fisheries Aquatic Sciences and Technology located in Barangay San Roque Mercedes, Camarines Norte. These exercises gave a descriptive insight into the existing work procedures, feeding schedules and issues that the pond manager and caretaker face daily. The Business Process Modeling (BPM) was conducted on the information collected to visualize and analyze the existing structure of operations.

Objective 2 deals with the establishment of the information requirements to develop the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder, both primary and secondary data was employed. The Information Requirement Analysis (IRA) was supported with the help of stakeholder interviews and system observations.

In Objective 3 that entails identification and design of system features, secondary sources that included

scholarly journals, research publications, and online literature on aquaculture practices, automated feeding systems, and mobile management application were used. These sources assisted in developing the technical design to make sure that the designed system is adjusted to the modern standards of aquaculture and the use of the latest technologies. Finally, Objective 4 that will provide insight on whether the developed system is effective based on the following aspects: perceived usefulness, perceived ease of use, ease of learning, and satisfaction will be driven by the Technology Acceptance Model (TAM). The main source of data that will be used to evaluate the acceptability and usability of the system will be user feedback obtained after the field testing and stakeholder evaluation. With the combination of both data-driven and literature-based knowledge, the proponents at least made sure that the process of system development was empirically based, grounded in the context, and scientifically grounded.

Population of the Study

The population of this study is specifically centered on the College of Fisheries Aquatic Science and Technology (CFAST), located in Barangay San Roque, Mercedes, Camarines Norte. This includes the farm manager, a caretaker responsible for pond maintenance, feeding, and monitoring, and representatives from the Bureau of Fisheries and

Aquatic Resources (BFAR) who provide technical guidance and regulatory insights for aquaculture operations. These individuals represent the primary users, stakeholders, and domain experts essential to the development, testing, and evaluation of the proposed system.

A similar population profile was reflected in the study by Hossain et al. (2023), which explored traditional fish farming practices and their socioeconomic impacts. Their research involved pond owners and caretakers, highlighting their critical roles in pond management and daily operational activities necessary for sustainable aquaculture.

Casinillo and Serriño (2023) adopted a census method in their study involving 118 youth members of the 4-H Club in Southern Leyte for data collection. Similarly, this study will employ census sampling—appropriate for small populations where every individual is included. The entire population consists of the CFAST fish farm manager, the caretaker, and selected BFAR personnel relevant to aquaculture management and advisory services. To ensure accurate and meaningful data collection, a series of interviews, observations, and system testing sessions will be conducted with the participants' consent. These procedures aim to gather information regarding current fish pond management practices, feeding routines, operational challenges, and opportunities for system enhancement.

Data Collection Method

The research team used qualitative and observational data collection techniques centered on the operations of the fish pond of the client in order to ascertain that the system development was founded on the practical operations.

First, the identification of the current business process of fish pond management was done by conducting semi-structured interviews with the fish pond manager and caretaker. These interviews gave an understanding of their daily routine such as feeding fish and overseeing it, choosing a feed formula and managing the pond in general. Moreover, observations on the spot were recorded to capture the reality of the working process, feeding schedules, mortality monitoring, and maintenance processes to have a full picture of the existing operation processes and difficulties experienced by the pond employees.

As a next step, the collected information from interviews and observations was analyzed to determine the necessary system inputs, outputs, and data flows required for designing the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder. Since no formal or complete manual records were available at the fish farm, the researchers relied primarily on the insights gathered from stakeholders and on concepts, formulas, and best practices derived from the Review of Related Literature (RRL). These sources served as the basis for

identifying the system's functional requirements, feeding computations, and operational processes. Despite the absence of actual documented logs, the team utilized the available qualitative information to develop the Information Requirements Analysis (IRA), ensuring that the system design accurately reflects both practical farm operations and established aquaculture standards. Based on this, the design and identification of the features of the system was informed by the insights gathered. The specification of functional requirements and user interface design of both the mobile application and the IoT-powered feeder were selected through interviews and observations. In addition, the secondary sources such as the scholarly journals, research publications, and literature on aquaculture practices and automated feeding systems were also referred to so that the design of the system would be in tandem with the current standards and technological advances in the management of aquaculture practices.

Lastly, the functionality of the developed system was tested by feedback sessions and the usability testing was done after the initial prototype was developed. These assessments were informed by the Technology Acceptance Model (TAM) to determine usefulness, satisfaction, ease of use, and ease of learning so that the formulated application and the IoT-driven feeder met the expectations and demands of operations of the client.

Data Analysis

The researchers will use both qualitative and quantitative methods to analyze the data in a more thorough way. Through qualitative analysis, the researchers will gather and interpret non-numerical information to better understand the real-life experiences of users, the challenges they faced, and how they were actually used in managing ponds that attributes in research question 1 and 2. This helped provide deeper insight into the day-to-day operations of the fish farm.

In contrast, the analysis of quantitative data generates numerical scores for evaluating the usability of the application, providing a tangible measure of the Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder's effectiveness. This method offers objective metrics that can be clearly measured and compared. When combined with qualitative analysis, it provides subjective insights that help interpret user experiences. Together, this mixed-method approach ensures a balanced and reliable evaluation of the system.

The researchers used the Likert Scale, as shown below in Table 2, to systematically quantify, collect, and analyze numerical data related to the usability of the Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder. This measurement approach enabled respondents to express their level of agreement or satisfaction across

predefined criteria, allowing subjective user perceptions to be translated into measurable values. The resulting data provided a structured basis for evaluating system effectiveness, ease of use, and overall user acceptance, supporting a more objective and reliable usability assessment.

Table 2
Likert Scale

Scale	Interval	Verbal Interpretation
5	4.21-5.00	Strongly Agree
4	3.41-4.20	Agree
3	2.61-3.4.20	Neutral
2	1.81-2.60	Disagree
1	1.00-1.80	Strongly Disagree

Statistical Treatment of Data

The weighted mean approach was applied to assess and extract the respondents' satisfaction insight with the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder. Various important factors were applied in this method such as perceived usefulness, perceived ease of use, attitude toward use or satisfaction of the use, and ease of learning all of which contributes to the performance overall of the system. The evaluation of the system and device's performance is quantified with the helped of the weighted mean from the respondents'

perceptions. This is a valuable statistical tool for the different levels of importance on various observations. Preferences of the users are analyzed by using this technique which emphasized the most significant factors reflecting them accurately.

The focus of this evaluation will be on assessing the usability of PONDIMATE: Fish Pond Management Mobile Application with IOT-Based Automatic Fish Feeder from the target users' perspective. The complexity of the user preferences is captured using the weighted mean, highlighting the most influence satisfaction in the criteria. Prioritizing the most impactful aspects of the simulation.

Hence, the calculation formula of the weighted mean is:

$$\bar{x} = \frac{\sum(wx)}{N}$$

where:

$\bar{x}$ = weighted mean

$\sum$ = weighted mean

W = weight

X = frequency

N = total number of respondents

The researchers provided quantifiable, detailed, and unbiased analyzation of the effectiveness of how the PONDIMATE application and IoT-based powered fish feeder

meets the needs and expectation of the users through this statistical technique. Key areas of the project were highlighted in the findings where it excels and where improvements can be applied for user experience to further enhance. Moreover, user preferences and specific challenges is identified in the analysis, offering optimization of the system's functionality and ensure engagement on the valuable insights. Thus, future updates and refinements to better align the application and device with user needs is informed with these results.

Ethical Consideration

The development of PONDIMATE is guided in adherence to ethical standards to ensure professionalism, transparency, and mutual respect between the researchers and the client. Aligned with this, formal request letters for information gathering and consultation were prepared and signed by the researchers and the Capstone Project Research Adviser. These documents were submitted to and approved by the client at the start of the pre- development phase.

Throughout the project's duration, the proponents prioritized the client's schedule, availability, and feedback. Consultation sessions and system assessments were scheduled with advance notice, and all interactions were documented, and to ensure clarity and accountability. Feedback was treated with importance and was promptly incorporated into the development process whenever

applicable. Moreover, the researchers maintained open and continuous communication with the client to align all project decisions with the client's goals and expectations. Transparency, respect for client input, and commitment to ethical collaboration were consistently upheld throughout the pre-development of the project.

The study upholds strict ethical standards, emphasizing the avoidance of plagiarism by properly citing all literature, theories, tools, and related works in APA 7th edition format. Ethical use of Artificial Intelligence (AI) is prioritized, addressing issues of privacy, bias, transparency, and accountability. Personal data are anonymized and protected with security frameworks, while the AI model is tested for biases to ensure fairness. Transparent documentation of data sources, algorithms, and decision-making supports verification, and human oversight ensures accountability. Guided by the principle of non-maleficence, the research restricts AI use to beneficial purposes, maintaining integrity and promoting responsible AI practices in research.

Chapter 4

RESULTS AND DISCUSSION

This chapter presents the results of the study focusing on the usability and effectiveness of the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder. The data collected from respondents were analyzed quantitatively to measure user satisfaction and system performance in fish farming operations.

Existing Business Process of CFAST Aquaculture Farm

After consultations and interview with the clients, the researchers come up with the existing business process of the client shown in Figure 4.

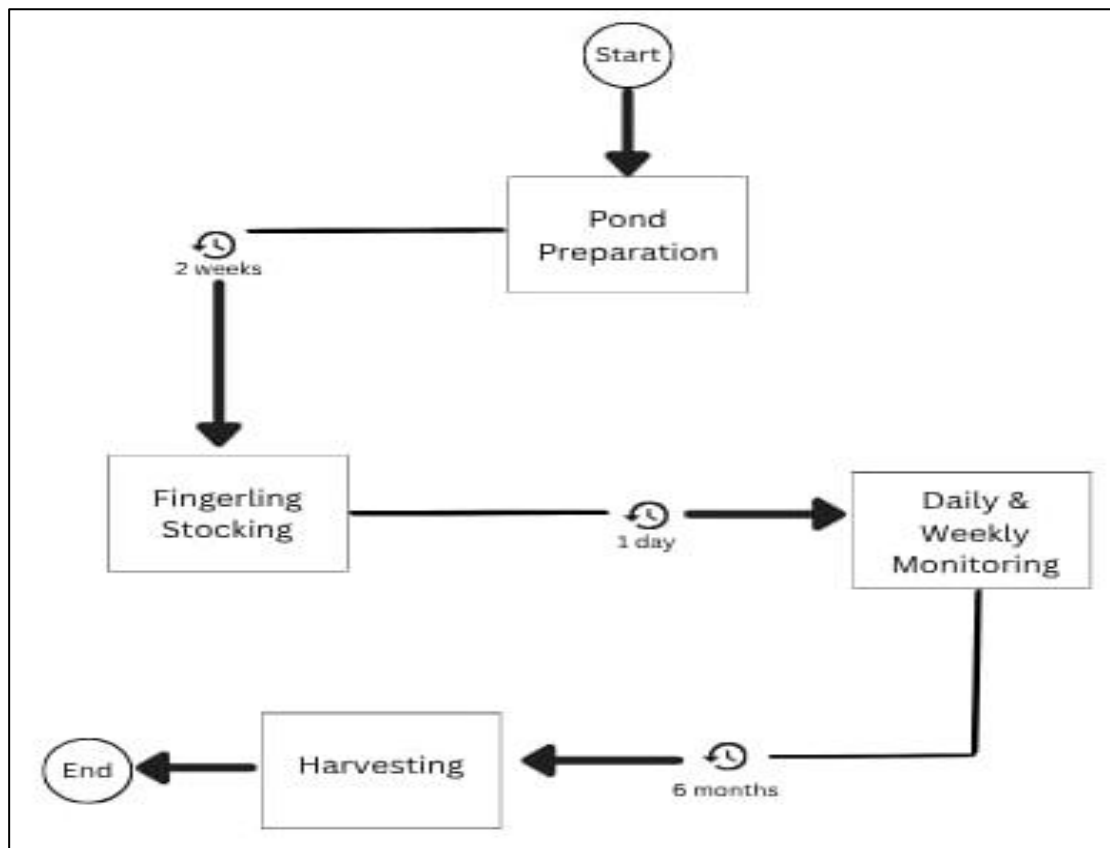


Figure 4. Existing Business Process Model of CFAST

The CFAST Aquaculture Farm business process is based on the semi-intensive farming system which is manual, labor-intensive, and needs tight integration between the management and the caretakers to maintain the production efficiency.

The pond preparation lasts for two weeks. Once prepared, the fingerlings are stocked, usually done early morning or late afternoon to reduce stress of fish and enhance a higher rate of survival. The next stage is the feeding management stage which lasts for six months until the time of harvesting.

Feeding is done twice a day (8-9 a.m. and 3-5 p.m.) with feed types changing according to the life stage of the fish: fry mash feed is used during the first month (blind feeding), crumble feed during the second month, and grower and finisher feeds from the third to the sixth month.

Sampling is conducted every 15 days after the blind feeding stage to determine the Daily Feed Ration (DFR), which dictates the amount of feed given to the pond.

Water quality parameters (pH, dissolved oxygen, temperature, and salinity) is monitored daily, while maintenance (vegetation control and dike repairs) is done monthly.

Harvesting tilapia and milkfish takes up to six months with two to four workers. After harvesting, the pond is emptied for a new cycle. Provided below in Table 3 is a

comprehensive list of activities for the whole production:

Table 3

Activities in Fish Pond Production

Day	Activity
1	Start Pond Preparation, Initial water flushed out
2	Close pond gate and apply organic tea seed powder (200 ppm)
3	Lime application to condition pond soil
4–9	Waiting period for lime effect
10	Flush out excess lime
11	Add water and apply fertilizer
12–14	Wait for fertilizer absorption
15	Stocking of fingerlings
16–44	Start blind feeding of fry mash to fingerlings
45	First sampling to determine DFR; Change feed from fry mash to crumble
46–59	Feed fingerlings based on computed DFR
60	Second sampling
61–74	Feed fingerlings based on updated DFR
75	Third sampling
76–89	Change feed from crumble to grower; Feed based on updated DFR
90	Fourth sampling
91–104	Feed fingerlings based on updated DFR
105	Fifth sampling
106–119	Change feed from grower to finisher; Feed based on updated DFR
120	Sixth sampling
121–134	Feed fingerlings based on updated DFR
135	Seventh sampling
136–149	Feed fingerlings based on updated DFR
150	Eighth sampling
151–164	Feed fingerlings based on updated DFR
165	Ninth sampling
166–179	Feed fingerlings based on updated DFR
180	Last sampling
181–193	Feed fingerlings based on updated DFR
194	Harvest day

Although effective, the whole process at CFAST is manual

since there will be handwritten records covering all the six ponds. This renders the process of monitoring activities such as feeding management, water quality recording and harvest tracking as time consuming and susceptible to human error. The intensity of such workflow is a labor-intensive process, which necessitates data management tools and technology to help to augment accuracy, lessen workload, and general efficiency of the farm.

Information Requirement Analysis

In this section, the information required on developing the application and IOT-powered fish feeder automation is define through utilizing Information Requirement Analysis (IRA). A visual representation of the Information Requirement is provided in Figure 5.

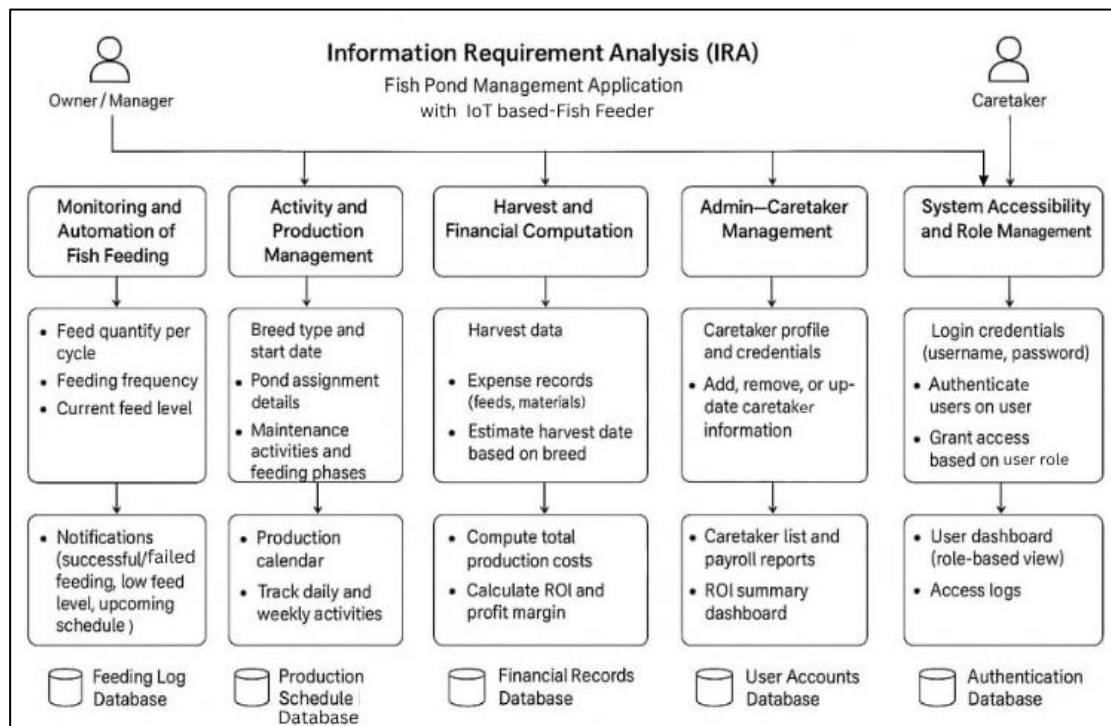


Figure 5. Information Requirement Analysis Diagram

The development of PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder was guided by a thorough understanding of the essential information requirements of both pond owners and caretakers. These requirements formed the foundation for the system's functionality, data processing, and user interface design. During the analysis phase, several key information needs were identified.

First, for monitoring and automation of fish feeding, the system must manage efficient and consistent feeding schedules by processing information such as feed quantity per cycle, feeding frequency, and current feed levels. The daily feeding rate is also calculated, every 15 days the client noted that they practice sampling of fish weight between at least 3 fish, the formula for calculating the Daily Feeding Rate of fish is based from Tilapia Production Guide by the Bureau of Fisheries and Aquatic Resources (2020), it is noted that feeding rates vary by fish growth stage, utilizing frymash, crumble, grower, and finisher feeds. During sampling, the Average Body Weight (ABW) of tilapia is measured every 15 days. ABW is calculated using the formula:

$$\text{Average Body Weight (ABW)} = \frac{\text{Total Weight of Sampled Stocks}}{\text{No. of Sample Stocks}}$$

This value represents the mean weight of the fish at a given sampling period and is required for determining the

Daily Feeding Rate (DFR). The formula for calculating the Daily Feeding Rate of fish is based on the *Tilapia Production Guide* by the Bureau of Fisheries and Aquatic Resources (2020), and is expressed as:

$$\text{Daily Feeding Rate (DFR)} = \text{No. of stocks} \times \text{SR\%} \times \text{ABW} \times \text{FR\%}$$

where:

No. of stocks = total number of stocks

SR = survival rate

ABW = average body weight

FR = feeding rate

In this equation, the Total Number of Stocks denotes the entire population of tilapia being cultured within the pond. The Average Body Weight (ABW) refers to the mean mass of individual fish at a given life stage, typically expressed in grams, and serves as a measure of biomass for estimating feed measurements. Meanwhile, the Percentage of Average Body Weight (%ABW) signifies the proportion of the fish's body weight that should be supplied as feed per day, which varies according to life stage. For instance, higher feeding rates are applied during the fingerling stage to support rapid growth, while lower percentages are used in later stages, such as the juvenile and adult phases, as metabolic and nutritional demands decrease. Together, these variables enable a precise estimation of daily feed requirements, ensuring efficient resource utilization and

promoting optimal growth performance within the aquaculture system.

The life cycle of both tilapia and bangus is also calculated. According to FAO (2020), *Oreochromis niloticus* or tilapia progresses through four major stages of growth. The cycle begins at the fingerling stage (0-60 days), where fish are stocked in grow-out ponds and require frequent feeding to ensure rapid development. This is followed by the juvenile stage (61-90 days), during which growth rates increase, and feeding remains consistent to support body mass gain. The sub-adult stage (91-180 days) focuses on adjusting feed quantities based on biomass and water quality conditions. Finally, by the adult or harvest stage (around 180 days), tilapia reach marketable size, completing the production cycle. See Figure 6.

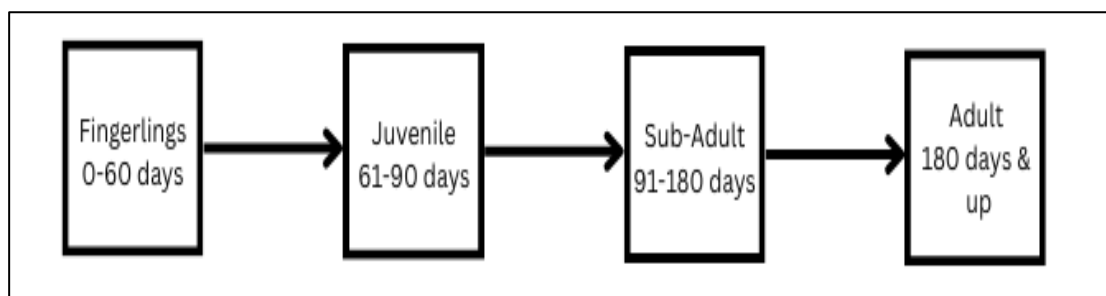


Figure 6. Life Stage of Tilapia

Similarly, shown in Figure 7, SEAFDEC (2021) outlines the corresponding life stages for milkfish (bangus), which also fit within a 180-day framework. The fingerling stage (0-70 days) involves initial stocking and high feeding frequency to stimulate growth. The juvenile stage (71-90 days) emphasizes continued feeding and monitoring of growth

and mortality. The sub-adult stage (91-180 days) requires feed adjustment according to biomass and close observation of health conditions. The cycle concludes with the adult or harvest stage (around 180 days), when milkfish reach optimal market size and harvesting takes place.

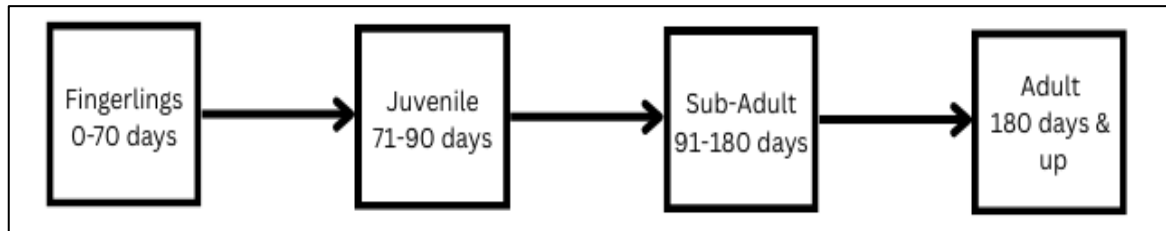


Figure 7. Life Stage of Bangus

Second, for activity and production management, the system should record and generate fish production schedules based on the breed and start date specified by the user. This includes storing breed details, pond assignments, daily and weekly maintenance tasks, and production phases such as pond preparation and feeding.

Additionally, it must estimate harvest dates and manage harvest operations on the caretaker's side, providing alerts for harvest readiness, as well as performing cost and revenue computations. To support financial decision-making, the system should capture expenses from pond setup to harvest, calculate production costs per pond, and project the return on investment (ROI).

As noted by Elliott (2022), ROI shows how much net gain is made from an investment relative to how much was spent to get that gain. The ROI can be represented as a percentage calculated from:

$$\text{Return of Investment (ROI)} = \frac{\text{Net Return}}{\text{Total Investment}} \times 100\%$$

where:

$$\text{Net Income} = \text{Gross Income} - \text{Total Cost}$$

$$\text{Total Investment} = \text{Fixed Costs} + \text{Variable Cost}$$

In this formula, Net Return (or Net Income) refers to the remaining profit after all expenses are deducted from the total revenue or Gross Income. On the other hand, Total Investment includes every cost required to operate the pond—both Fixed Costs (long-term expenses such as pond construction, equipment, and infrastructure) and Variable Costs (recurring expenses like fish feed, labor, utilities, and maintenance). For the estimation of ROI, feed and fertilizer costs are calculated based on the pond's area and stocking density, as these directly affect the amounts of inputs needed. Larger ponds or higher stocking densities require more feed and fertilizer to sustain fish growth, while smaller ponds or lower densities use proportionally less.

Furthermore, the system incorporates an admin-caretaker management feature that enables the owner to oversee caretaker employment through functions such as adding or removing caretakers. Lastly, in terms of user roles and system accessibility, the application supports online mode and distinguishes between two user types: the owner or manager, who has full system access, and the

caretaker, who has limited access focused on feeding schedules, weekly pond activities, and production tasks.

Fish Pond Management Application Features

Fishpond Management Application is a cohesive application that will be used together with an IoT-based fish feeding device to allow users to make evidence-based decisions on cost-efficiency, production management, and profitability.

Firstly, Figure 8 illustrates how user control ensures secure, organized, and structured system management through clearly defined access levels. Administrators are granted full control over all system functions and data, including monitoring farmgate and feeder prices, managing system configurations, and linking IoT devices to specific ponds for efficient oversight.

In contrast, caretakers are granted limited yet carefully defined access rights that are specifically tailored to their responsibilities within the fish pond management system. This controlled access enables them to carry out essential operational tasks such as daily monitoring of pond conditions, performing feeding routines, maintaining equipment, and ensuring the overall health and growth of the fish stock. By restricting access to only the functions necessary for their role, the system effectively minimizes the potential for unauthorized actions or accidental errors that could compromise data integrity or

pond operations. Furthermore, this role-based access approach not only enhances accountability by clearly delineating who is responsible for each activity but also supports efficient workflow management, allowing caretakers to focus on their duties without distraction. Overall, providing caretakers with role-specific access helps maintain the security, reliability, and smooth functioning of the fish pond management system, ensuring that all users can fulfill their responsibilities effectively while safeguarding operational continuity.

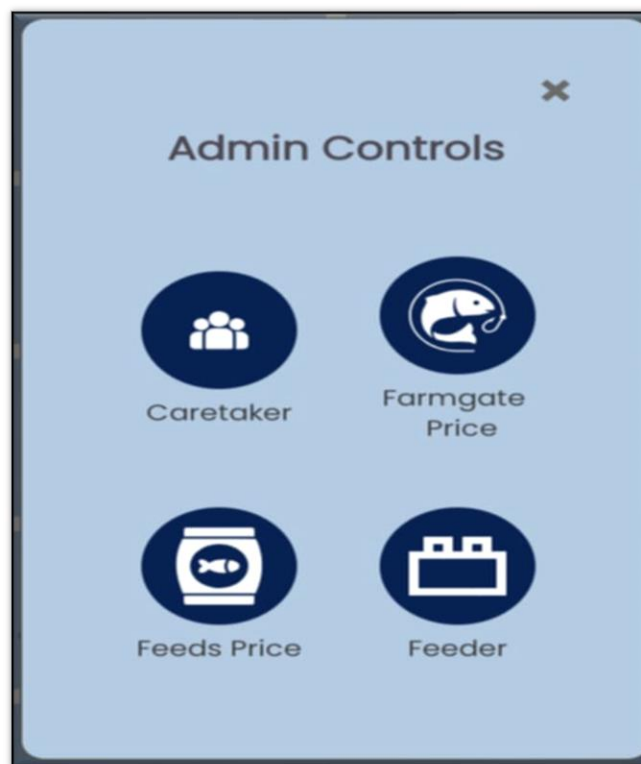


Figure 8. User Control

Moreover, Figure 9 highlights the grow-out cycle feature, which enables continuous and systematic monitoring of pond progress and the developmental stages of cultured fish. This feature records and analyzes key growth metrics over

time, allowing users to assess fish health, feeding efficiency, and overall pond conditions. By providing timely and accurate data, it supports early interventions to address potential issues, optimize fish performance, and reduce mortality rates. Ultimately, the grow-out cycle feature promotes evidence-based decision-making for adjusting feeding schedules, managing resources efficiently, and determining the most appropriate harvesting period.

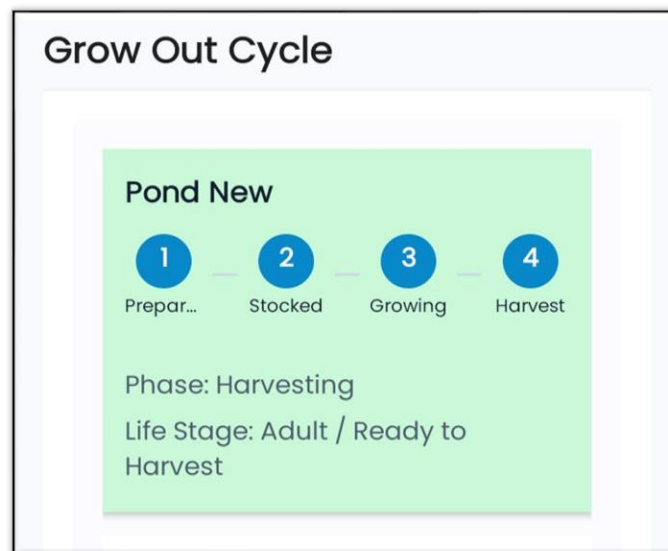


Figure 9. Grow-Out Cycle

Complementing this, Figure 10 presents the pond history feature, which generates a PDF report containing comprehensive records of a pond's operational data for future reference. This report consolidates information such as feeding activities, growth performance, maintenance logs, and notable events throughout the production cycle. By having an organized and downloadable record, both administrators and caretakers can easily analyze long-term

trends, identify recurring issues, and evaluate the effectiveness of past management practices. Consequently, this feature enhances traceability, supports accountability, and contributes to continuous improvement in production strategies and overall pond management.

POND REPORT

This report covers the culture period from 2025-10-03 to 2025-11-05
Generated on: November 5, 2025 10:52 AM

POND DETAILS

Pond Name: Aa
Pond Area (m²): 111
Total Stocks: 100
Mortality Rate: 5
Recommended Culture Period: 6 months
Species: Bangus
Farmgate Price: 40
Total Harvest: 95

FINGERLINGS

Description	Quantity	Cost per Unit	Amount
Bangus	100	0.50	50.00
			Total: 50.00

ACCUMULATED FEED COST

Description	Amount
Blind Feeding	921.10
Sampling Feed Cost	2524.95
Total Feed Cost (Accumulated + Blind):	3446.05

OTHER EXPENSES

Type	Description	Amount
maintenance	Miscellaneous Expenses	1021.00
maintenance	Labor	1500.00
		Total: 2521.00

CARETAKER SALARY

Description	Amount
Caretaker Salary (4 months)	80000.00
Total Salary: 80000.00	

TOTAL CAPITAL / EXPENSES: 86017.05

--- End of Report ---

Figure 10. Pond Report

In addition, Figure 11, the calendar of activities, organizes and schedules all pond-related tasks, ensuring structured and timely management of daily and periodic operations. Users can visualize upcoming maintenance,

feeding, and monitoring activities, which reduces the risk of missed tasks and streamlines coordination among staff. This feature also allows users to track completed and pending activities, promoting accountability and efficient workflow management. By centralizing all scheduled tasks in a single interface, the system enhances planning accuracy and supports better decision-making. Furthermore, the calendar enables the farm manager to anticipate resource needs ahead of time, contributing to more sustainable and well-coordinated fishpond operations.

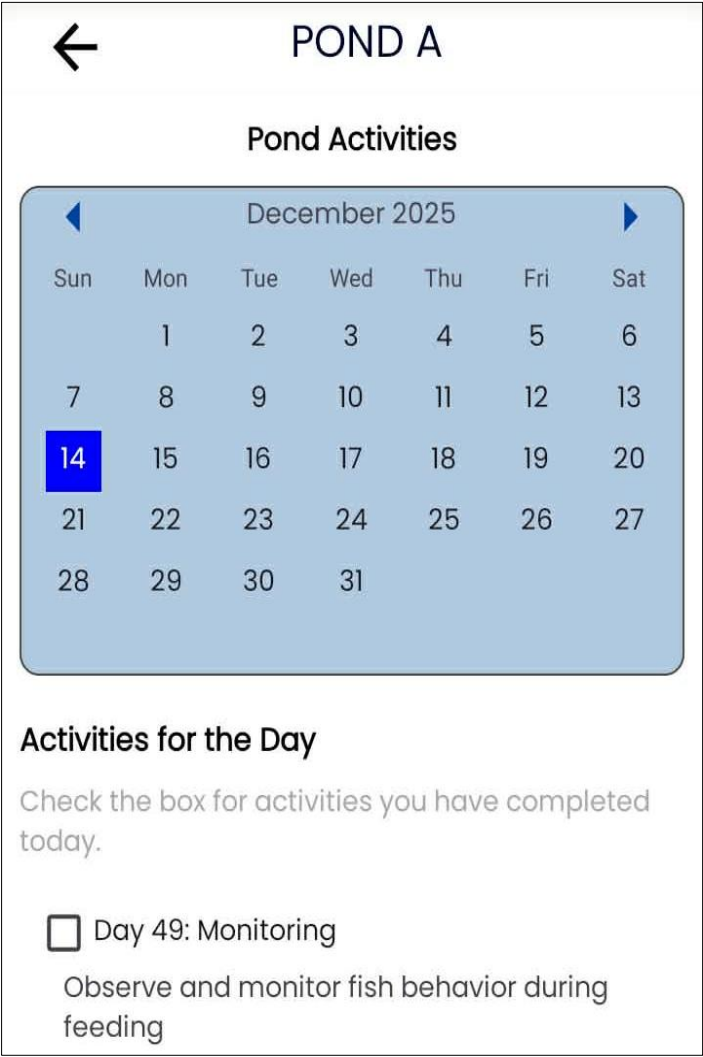


Figure 11. Calendar of Activities

Financial management is further supported through the production cost summary in Figure 12, which provides a detailed overview of all expenses associated with each pond. By aggregating costs related to feeds, labor, maintenance, and equipment. This feature facilitates informed budgeting, cost control, and resource allocation decisions. Overall, this summary serves as a valuable tool for improving financial transparency.

Production Cost Summary	
Category	Amount
Fingerlings	₱50.00
Accumulated Feed Costs	
Feed Cost for Blind Feeding	₱921.10
Sampling Feed Cost (every succeeding cycle)	₱2524.95
Other Expenses	₱2521.00
List of Expenses	
Miscellaneous Expenses	₱1021.00
Labor	₱1500.00
Salary	₱12000.00
Total	₱18017.05

Figure 12. Production Cost Summary

Consequently, in Figure 13, the return of investment analysis evaluates profitability by comparing total

production costs against generated revenue. This allows users to quantify economic performance and make strategic decisions regarding pond management and expansion, ultimately supporting long-term sustainability. By presenting financial outcomes in a clear and visual format, the system helps users quickly determine whether current practices are yielding favorable returns.

PROJECTED ROI BREAKDOWN	
Fingerlings Stocked:	100 pcs
Mortality Rate:	5.00%
Survival Rate:	95.00%
PROJECTED SALES	
Harvest Weight:	23.75 kg
Farm-Gate Price:	₱110.00/kg
Gross Revenue:	₱2612.50
FIXED COSTS (6 months)	
Fingerlings Cost:	₱50.00
Caretaker Salary:	₱12000.00
Supplies & Materials:	₱6000.00
Misc. Expenses:	₱6000.00
Harvesting (2%):	₱52.25
Caretaker Incentives (5%):	₱130.63
Other Expenses (2%):	₱52.25
VARIABLE COSTS	
Feeds (60%):	₱39445.25
Fertilizer (3.06%):	₱2011.71
TOTAL & ROI	
TOTAL EXPENSES:	₱65742.08
ESTIMATED ROI:	-96.03%
OK	

Figure 13. Return of Investment

Furthermore, Figure 14, the feeding schedule feature

provides structured guidance for the timing, frequency, and quantity of feed, aligning feeding activities with fish growth stages and pond conditions. This improves growth performance and overall production efficiency. This feature also enables accurate monitoring of feed usage and cost, supporting effective resource management and financial planning within the aquaculture system. Additionally, it helps standardize feeding practices, ensuring consistency across different production cycles.

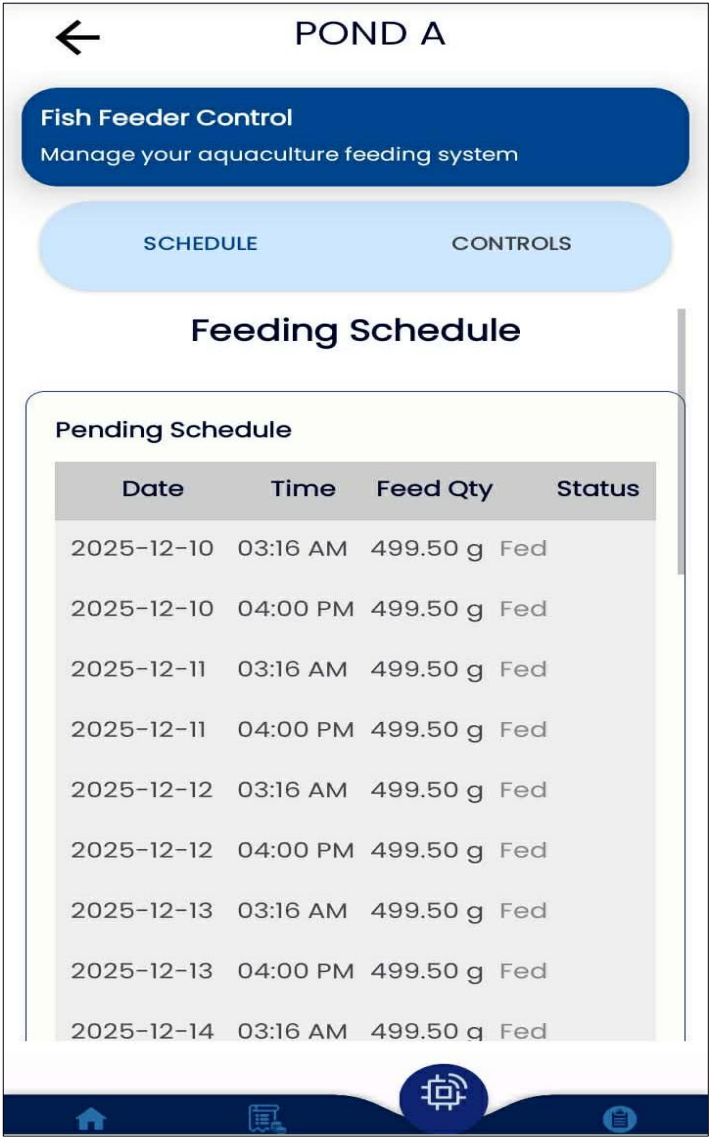


Figure 14. Feeding Schedule

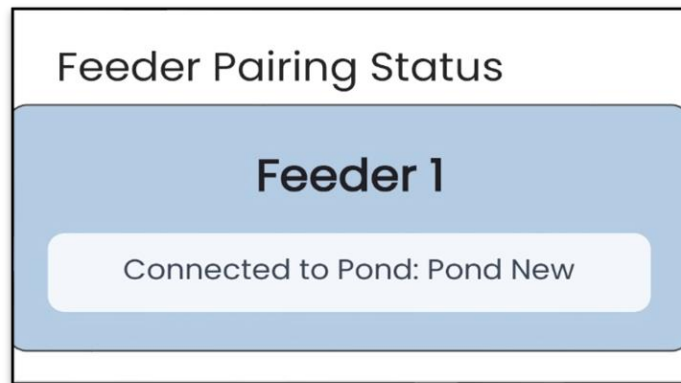


Figure 15. IOT Connection Status

Finally, Figure 15 shows the IoT connection status, displaying the unique feeder ID and the pond it is connected to.

The Fishpond Management Application in general is a combination of automation, monitoring and analysis tools that help to streamline the operations of aquaculture and improve the process of decision making as well as to guarantee the sustainable productivity of the fish farming systems.

Automated IOT-Based Fish Feeder Features

Designed to automate and simplify the aquaculture of the College of Fisheries Aquatic Sciences and Technology, the IoT-based fish feeding device combine components with smart features. Ensuring that fish are properly fed, monitored resources, guaranteeing the specific role that the components play in building the device. The IoT-based automatic fish feeder represented in Figure 16 is an advanced aquaculture technology designed to automate and optimize the feeding process for fish ponds. This device integrates

multiple functional components to ensure precise and efficient feed distribution, while simultaneously reducing labor requirements and minimizing feed wastage.

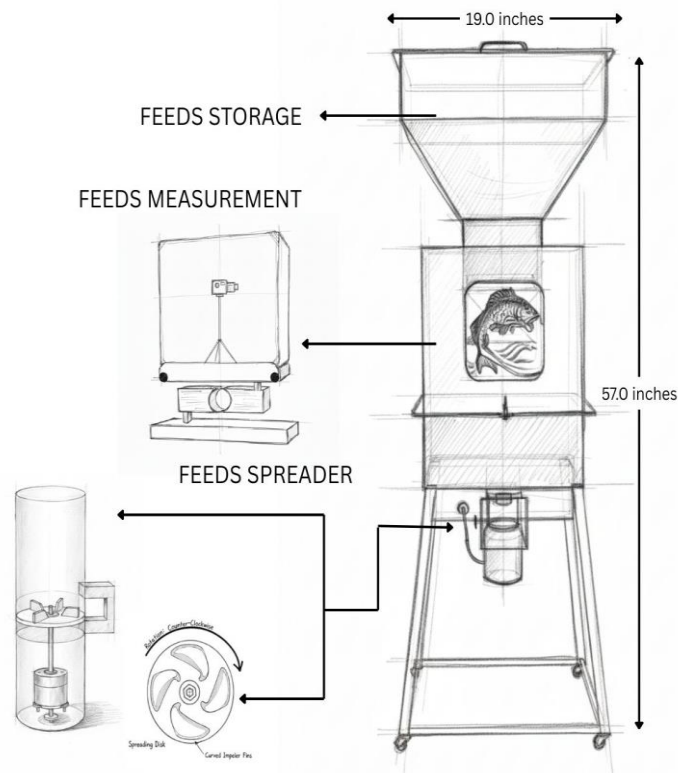


Figure 16. 3D Design of IOT-Based Fish Feeder

The feed storage unit, located at the upper compartment of the device, is designed to hold adequate quantities of feed, thereby enabling multiple automated feeding cycles without frequent manual replenishment. Directly below it is the feed measurement mechanism, it regulates the quantity of feed dispensed per cycle, ensuring that fish receive portions calibrated according to predefined daily feeding rate. Subsequently, the feeds are transferred to the spreading mechanism, which distributes the feed across the pond surface. This component, equipped with a rotating disk, prevents localized overfeeding and promotes uniform consumption by all

fish within the pond. IoT integration enables remote monitoring and control of feeding, allowing users to schedule feedings, adjust portions, and track feeder activity for consistent operation without on-site personnel. The feeder, mounted on a 19×57-inch stable frame, is designed for durability, easy maintenance, and proper feed dispersal. The device shown in Figure 17 is designed to automate the process of scheduled and on-demand feeding and monitor remaining feed levels. The materials used for the feeder is ensured to be weather-proof. These features collectively enhance automation, precision, and management efficiency in aquaculture operations.



Figure 17. Final Version of IOT-Based Fish Feeder

The scheduled feeding function allows automatic

dispensing of feed at predetermined times without human intervention. This process is precisely managed by the DS3231 Real-Time Clock (RTC) module to maintain accurate timing, while the Arduino Mega 2560 controls the feeding mechanism by activating either a DC gear motor via the L298N motor driver for larger feed systems or a micro servo motor for smaller dispensers.

A stable 12V power supply is maintained through LM2596 buck converters, ensuring consistent and reliable operation essential for optimal fish growth and pond management. The on-demand feeding feature enables flexible feeding schedules through the ESP8266 NodeMCU v3, which connects to Wi-Fi and allows feeding commands to be sent directly from the Fishpond Management Mobile Application.

Upon receiving a command, the Arduino Mega 2560 triggers the appropriate motor or servo to release feed as needed, allowing immediate adjustments to feeding routines.

Additionally, feed level monitoring is achieved using a load cell sensor connected to an HX711 amplifier, which continuously measures the hopper's weight. The Arduino Mega 2560 interprets these readings to calculate remaining feed quantities, displaying the results on an LCD 2004a for local monitoring. When feed levels fall below a defined threshold, the system generates alerts to prevent feeding interruptions and support efficient resource management.

The Arduino Mega 2560 records feeding activities and

sensor data, while the NodeMCU transmits this information to the mobile application once internet connectivity is available. These integrated notifications improve responsiveness, reduce manual monitoring, and enhance overall oversight of aquaculture operations.

Evaluation Results of Metrics and Criteria for the Implemented System

The evaluation of the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder assessed how effectively the system meets user needs and expectations. This section presents results based on metrics aligned with the Technology Acceptance Model (TAM), including Perceived Usefulness, Perceived Ease of Use, Ease of Learning, and User Satisfaction. Data were collected from thirty (30) respondents, including the main client, and analyzed using the weighted mean method.

In Table 4, Statements 1 ("The application helps me manage my fish pond more efficiently") and 4 ("The system improves the accuracy of my pond monitoring activities") obtained the highest weighted mean of 4.70, interpreted as Strongly Agree, indicating high recognition of the system's efficiency and accuracy. Meanwhile, Statements 2, 3, and 5 recorded the lowest weighted mean of 4.63, also interpreted as Strongly Agree, showing that respondents still strongly perceive the system as useful in reducing feeding errors, improving decision-making, and enhancing productivity. The

pooled weighted mean of 4.66, interpreted as Strongly Agree confirms the high perceived usefulness of the application and device.

Table 4

Perceived Usefulness (PU) of the Application and Device

Statement	Weighted Mean	Interpretation
1. The application helps me manage my fish pond more efficiently.	4.70	Strongly Agree
2. The IoT feeder reduces errors in daily feeding.	4.63	Strongly Agree
3. The system allows me to make better decisions in pond management.	4.63	Strongly Agree
4. The system improves the accuracy of my pond monitoring activities.	4.70	Strongly Agree
5. Using the application enhances the overall productivity of my fish pond.	4.63	Strongly Agree
Pooled Weighted Mean	4.66	Strongly Agree

In Table 5, Statements 3 ("Navigating through the system requires little effort") and 5 ("I can use the system without needing frequent assistance") obtained the highest weighted mean of 4.60, interpreted as Strongly Agree, indicating that respondents find the system easy to navigate and use independently. Meanwhile, Statement 1

("The application is easy to use") recorded the lowest weighted mean of 4.30, interpreted as Agree, suggesting that while the application is generally easy to use, there is still room for improvement. The pooled weighted mean of 4.50, interpreted as Strongly Agree, confirms that the application and device are perceived as easy to use.

Table 5
Perceived Ease of Use (PEOU) of the Application and Device

Statement	Weighted Mean	Interpretation
1. The application is easy to use.	4.30	Agree
2. The IoT feeder is simple to operate and setup.	4.50	Agree
3. Navigating through the system requires little effort.	4.60	Strongly Agree
4. The features of the application are easy to understand.	4.50	Agree
5. I can use the system without needing frequent assistance.	4.60	Strongly Agree
Pooled Weighted Mean	4.50	Strongly Agree

In Table 6, Statements 3 ("It was easy for me to become skillful in using the application") and 4 ("Learning to operate the IoT feeder was straightforward") obtained the highest weighted mean of 4.73, interpreted as Strongly Agree, indicating that respondents found the system and device easy to master. Meanwhile, Statement 1 ("I quickly

learned how to use the application”) recorded the lowest weighted mean of 4.03, interpreted as Agree, suggesting that initial learning may require some adjustment for users. The pooled weighted mean of 4.57, interpreted as Strongly Agree, confirms that the application and device are easy to learn.

Table 6

Ease of Learning (EL) of the Application and Device		
Statement	Weighted Mean	Interpretation
1. I quickly learned how to use the application.	4.03	Agree
2. Instructions and guidance in the system are clear and helpful.	4.70	Strongly Agree
3. It was easy for me to become skillful in using the application.	4.73	Strongly Agree
4. Learning to operate the IoT feeder was straightforward.	4.73	Strongly Agree
5. I can learn new features of the system without difficulty.	4.67	Strongly Agree
Pooled Weighted Mean	4.57	Strongly Agree

In Table 7, Statement 4 (“The system meets my expectations as a pond caretaker/owner”) obtained the highest weighted mean of 4.80, interpreted as Strongly Agree, indicating that the system effectively meets users’ expectations. Meanwhile, Statement 1 (“I am satisfied with the features of the application”) recorded the lowest

weighted mean of 4.53, which is also interpreted as Strongly Agree, showing that respondents are still highly satisfied with the application features. The pooled weighted mean of 4.70, interpreted as Strongly Agree, confirms a very high level of user satisfaction with the application and device.

Table 7

Satisfaction (SAT) of the Application and Device

Statement	Weighted Mean	Interpretation
1. I am satisfied with the features of the application.	4.53	Strongly Agree
2. I am satisfied with the IoT-based automatic feeder	4.67	Strongly Agree
3. Overall, I am satisfied using the system in my pond management.	4.77	Strongly Agree
4. The system meets my expectations as a pond caretaker/owner.	4.80	Strongly Agree
5. I would recommend this system to other pond managers.	4.73	Strongly Agree
Pooled Weighted Mean	4.70	Strongly Agree

The overall evaluation of the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder in Table 8 indicates a high level of acceptance and positive perception among the respondents.

In Table 8, Statement 4 (Satisfaction) obtained the highest weighted mean of 4.80, interpreted as Strongly Agree, indicating a very high level of user satisfaction with the application and device. Meanwhile, Statement 2 (Perceived Ease of Use) recorded the lowest weighted mean of 4.50, which is still interpreted as Strongly Agree, suggesting that the system is generally easy to use. The pooled weighted mean of 4.60, interpreted as Strongly Agree, confirms the overall positive evaluation and acceptance of the application and device.

Table 8

Overall Result of the Application and Device

Statement	Weighted Mean	Interpretation
1. Perceived Usefulness	4.66	Strongly Agree
2. Perceived Ease of Use	4.50	Strongly Agree
3. Ease of Learning	4.77	Strongly Agree
4. Satisfaction	4.80	Strongly Agree
Pooled Weighted Mean	4.60	Strongly Agree

Respondents strongly agreed that the application and IoT feeder enhance the efficiency and accuracy of fish pond management, support better decision-making, and improve overall productivity, reflecting a high perceived usefulness. The system was also rated as user-friendly, with intuitive navigation and straightforward operation of both the application and the IoT device, demonstrating

strong perceived ease of use. In terms of learning, users reported that the system is easy to understand and that they were able to quickly become proficient, highlighting its low learning curve and effective instructional support. Finally, the respondents expressed high satisfaction, affirming that the system meets their expectations, enhances their daily pond management, and would be recommended to others.

Overall, these findings suggest that the PONDIMATE system successfully delivers practical, reliable, and user-friendly solutions for fish pond management. The consistently high ratings across all constructs indicate that the application and IoT-based automatic feeder are effective, efficient, and well-received by users, providing a strong foundation for further development, optimization, and wider adoption.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter presented the conclusions and recommendations that are relevant to the study.

Conclusions

The conclusions which are made following the results of the study are as follows:

1. CFAST Aquaculture Farm uses a semi-intensive, manual system that relies on handwritten records, making operations labor-intensive and prone to errors. Adopting data management tools would improve accuracy, reduce workload, and increase overall efficiency;

2. The Information Requirement Analysis defines the key data needed for the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder, including feeding schedules, feed calculations, growth tracking, activity management, costs, and ROI, ensuring efficient pond operations;

3. The PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder streamline aquaculture by integrating grow-out tracking, activity calendars, reports, cost and ROI summaries, feeding schedules, and IoT controls. The automated feeder handles scheduled and on-demand feeding, monitors feed levels, and reduces labor and errors, providing real-time data that improves productivity and supports sustainable pond

management; and

4. The evaluation of PONDIMATE shows strong user acceptance, with all TAM indicators receiving high scores and an overall mean of 4.60 (Strongly Agree). Users found the app and IoT feeder efficient, accurate, easy to use, and effective for pond management, confirming the system's practicality, reliability, and user-friendliness.

Recommendations

Recommendations based on findings and conclusions of the study are as follows:

1. Implement the system and automate pond operations to eliminate the manual and labor-intensive processes currently used in CFAST Aquaculture Farm. Integrating the PONDIMATE: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder into day-to-day pond management will enhance productivity, improve accuracy in feeding, monitoring, and record-keeping, and ensure consistent operational efficiency;

2. Incorporate a water quality monitoring feature to track critical parameters such as pH, dissolved oxygen, temperature, and salinity in real-time. This addition would allow caretakers and administrators to maintain optimal pond conditions, detect abnormalities early, and make informed decisions to improve fish health and production outcomes;

3. Enhance user convenience and accessibility by

integrating a solar panel system to ensure continuous power supply for IoT devices, and enabling offline functionality in the application to support pond management even in areas with unstable or limited internet access. Further, incorporate a highly accurate feed-dispensing mechanism to ensure precise feed measurement at all times. Finally, provide comprehensive user training for both administrators and caretakers prior to implementation to promote effective system adoption and usage; and

4. Ensure system reliability and continuous improvement through regular updates, refinement of features based on user feedback, and maintenance of both the mobile application and IoT feeder. This will enhance operational control, sustain user satisfaction, and support long-term sustainable aquaculture management.

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APPENDICES

Appendix A

Certificate of Editor



Republic of the Philippines
University of the Philippines Los Baños
COLLEGE OF ARTS AND SCIENCES
Los Baños, Laguna



EDITOR'S CERTIFICATION

This is to certify that the undergraduate thesis titled "PONDIMATE: FISH POND MANAGEMENT MOBILE APPLICATION WITH IOT-BASED AUTOMATIC FISH FEEDER," prepared by MACAPAGAL, RHYZELINE M., VILLARANDA, VERGEL B., LABRADOR, FRANCES FIONA P., and BABEJES, PAMELA JANE T., in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, is thoroughly reviewed and has completely undergone grammarian technique following the set of structural rules that govern the composition of sentences and words in English language.

Signed by:

MARIANNE C. GARCIA, LPT

Editor, License No. 2017016

Date:

December 17, 2025

Appendix B

Request Letter for Research Adviser

Spencer Saludes
Instructor I
College of Computing and Multimedia Studies
Camarines Norte State College

February 18, 2025

Subject: Request for Capstone and Research Advisor Approval

Dear Sir,

We are third-year BSIT students looking for an advisor for our capstone project and research. We would be honored if you would consider taking on this role.

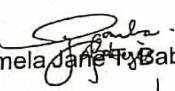
Our "PondPal: Fishpond Management System with IoT based Fish Feeder" project, an application designed to streamline and optimize fish farming operations. This system provides a platform for managing feeding schedules, tracking expenses, and estimating harvest times.

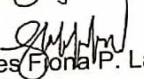
As our advisor, your insights and mentorship would be instrumental in refining our methodology, ensuring the rigor of our research, and helping us translate our findings into a well-structured final project. We understand the commitment involved and will ensure that we maintain regular communication and meet deadlines efficiently.

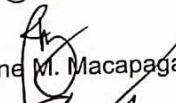
We would greatly appreciate the opportunity to discuss this further at your convenience. Please let us know if you would be open to advising us and if there are any additional requirements we should fulfill. Thank you for your time and consideration.

We look forward to your response.

Best regards,

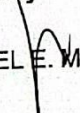

Pamela Jane T. Bajejes


Frances Fiona P. Labrador

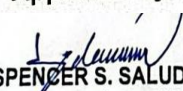

Rhyzeline M. Macapagal


Vergel B. Villaranda

Noted by:


DANIEL E. MALIGAT JR, DIT
Dean

Approved by:


SPENCER S. SALUDES
Capstone Project and Research Adviser

Appendix C

Request Letter for Client Approval

October 07, 2025

MARJORIE S. LUCAS, MSc
Instructor I
College of Fisheries, Aquatic Sciences, and Technology

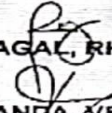
Greetings!

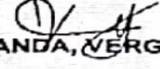
We are fourth-year students from Camarines Norte State College – College of Computing and Multimedia Studies under the Bachelor of Science in Information Technology (BSIT) program. This letter is submitted in line with our IT-121 Capstone Project and Research 1 course, which focuses on developing innovative software and IoT-integrated systems that address real-world community needs.

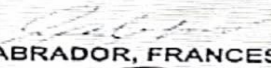
In connection with this, we respectfully request permission to conduct an interview with the manager or any staff member knowledgeable about the operations and management practices of the College of Fisheries, Aquatic Sciences, and Technology (CFAST) Aquaculture Farm. This interview is an essential part of our research study entitled **PondMate: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder**. Our objective is to understand the current business processes and farm management procedures in place, which will guide us in designing a system that is efficient, accurate, and relevant to your operations. The cooperation, expertise, and insights of the respective Manager, Caretaker, and Faculty will be highly valuable in completing this study and ensuring the development of a functional and beneficial digital system. We assure you that all gathered information will remain confidential, used exclusively by the proponents, and strictly for academic purposes.

We look forward to your favorable response and hope that this research will contribute to both the improvement of your aquaculture operations and the development of our academic work. Thank you!

Sincerely yours,

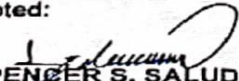

MACAPAGAL, RHYZELINE M.


VILLARANDA, VERGEL B.


LABRADOR, FRANCES FIONA P.


BABEJES, PAMELA JANE T.

Noted:


SPENCER S. SALUDES
Capstone Project and Research Adviser


DANIEL MALIGAT JR., DIT
Capstone Project and Research Adviser

Approved:


MARJORIE S. LUCAS, MSc
Instructor I
College of Fisheries, Aquatic Sciences, and Technology

Appendix D

Letter for Pre-Final Defense

September 30, 2025

DANIEL E. MALIGAT JR., DIT
Capstone Project and Research 2 Professor
College of Computing and Multimedia Studies

Sir:

In partial fulfillment of the requirement in the Bachelor of Science in Information Technology program, the undersigned conducted a Capstone Project entitled "**PondPal: Fishpond Management System with IOT-based Fish Feeder**".

In connection with this, the undersigned humbly request the following CCMS faculty as members of our panel committee. Apart from that, we would like to schedule our pre-final defense on October 2, 2025, at est 8:00 AM to 12:00 PM.

Thank you very much!

Panel Committee Members:

NORIANNE C. LAMADRID

Signature

Date

EDGAR BRYAN B. NICART, DIT

Signature

Date

Respectfully yours,

BABELLES, PAMELA JANE T.

LABRADOR, FRANCIS FIONA P.

MACAPAGAL, RHYZELINE M.

VILLARANDA, VERGEL B.

Noted:

SPENCER S. SALUDES, MIS
Capstone Project and Research Adviser

Appendix E

Request Letter for Evaluation

November 20, 2025

DANIEL MALIGAT., DIT
Dean, College of Computing and Multimedia Studies
Camarines Norte State College

Sir,

Greetings!

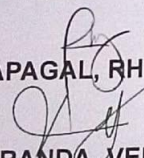
We are the fourth-year students pursuing a Bachelor of Science in Information Technology at Camarines Norte State College writing to humbly request your approval to conduct an evaluation of our Capstone Project entitled, "Pondmate: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder".

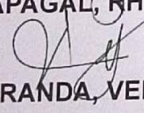
Pondmate is a mobile application to streamline aquaculture management through features such as feeding schedule management, operational cost monitoring, and harvest estimation. The system also integrates an IoT-controlled automatic fish feeder, allowing users to remotely set and manage feeding times through a mobile interface.

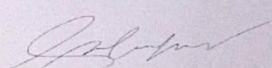
The evaluation will be conducted on November 20, 2025 onwards, utilizing the Technology Acceptance Model (TAM) questionnaire. The assessment will involve 30 respondents consisting fish pond owners, fish pond caretakers, faculties from Camarines Norte Fisheries, Aquatic Sciences, and Technology (CFAST), and related departments such as Bureau of Fisheries and Aquatic Resources (BFAR), and Municipal Agriculture Office (MAO). All responses will be confidential and will not be used outside this research in determining the system's level of acceptance.

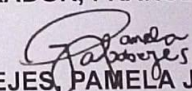
Thank you very much for your time and consideration.

Sincerely yours,

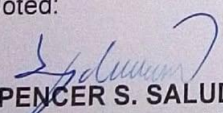

MACAPAGAL, RHYZELINE M.

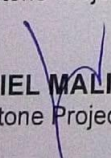

VILLARANDA, VERGEL B.


LABRADOR, FRANCES FIONA P.


BABEJES, PAMELA JANE T.

Noted:


SPENCER S. SALUDES
Capstone Project and Research Adviser


DANIEL MALIGAT JR., DIT
Capstone Project and Research Adviser

Appendix F

Request Letter for Evaluation in BFAR

November 19, 2025

Respondent Letter

Dear Respondents,

Greetings!

We are the fourth-year students pursuing a Bachelor of Science in Information Technology at Camarines Norte State College writing to humbly request your approval to conduct an evaluation of our Capstone Project entitled, "Pondmate: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder".

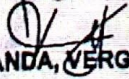
Pondmate is a mobile application to streamline aquaculture management through features such as feeding schedule management, operational cost monitoring, and harvest estimation. The system also integrates an IoT-controlled automatic fish feeder, allowing users to remotely set and manage feeding times through a mobile interface.

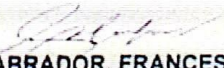
The evaluation will be conducted on November 20, 2025 onwards, utilizing the Technology Acceptance Model (TAM) questionnaire. The assessment will involve 30 respondents consisting fish pond owners, fish pond caretakers, faculties from Camarines Norte Fisheries, Aquatic Sciences, and Technology (CFAST), and related departments such as Bureau of Fisheries and Aquatic Resources (BFAR), and Municipal Agriculture Office (MAO). All responses will be confidential and will not be used outside this research in determining the system's level of acceptance.

Thank you very much for your time and consideration.

Sincerely yours,

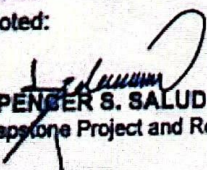

MACAPAGAL, RHYZELINE M.


VILLARANDA, VERGEL B.


LABRADOR, FRANCES FIONA P.


BABEJES, PAMELA JANE T.

Noted:


SPENCER S. SALUDES
Capstone Project and Research Adviser


DANIEL MALIGAT JR., DIT
Capstone Project and Research Adviser

BFAR
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RECEIVED
11/19/25

Appendix G

Letter for External Panel

December 9, 2025

MARJORIE S. LUCAS, MSc

Instructor I

College of Fisheries, Aquatic Sciences, and Technology

Sir,

We, the fourth-year Bachelor of Science in Information Technology students from Camarines Norte State College, are reaching out regarding our Capstone Project titled "PondMate: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder."

PondMate is a mobile application designed to support efficient and modernized fish pond operations. The system integrates IoT technology to automate feed dispensing, monitor pond conditions, and assist aquaculture personnel in managing daily tasks. By providing automated processes and organized record-keeping, PondMate aims to improve farm productivity, reduce manual labor, and enhance overall aquaculture management practices.

Furthermore, we would like to formally invite you to serve as an external panelist during our final defense, scheduled for December 10, 2025, at 3:00 – 4:00 P.M Your expertise, insights, and professional feedback will be invaluable in strengthening our project and ensuring its relevance and effectiveness in real-world aquaculture operations.

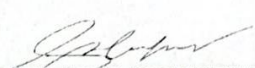
We sincerely hope you will consider our request. Your support would mean a great deal to us, and we look forward to your positive response.

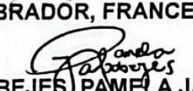
Thank you very much for your time and consideration.

Sincerely yours,

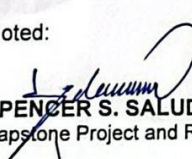

MACAPAGAL, RHYZELINE M.


VILLARANDA, VERGEL B.


LABRADOR, FRANCES FIONA P.


BABEJES, PAMELA JANE T.

Noted:

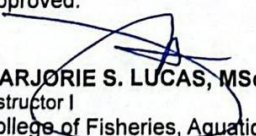

SPENCER S. SALUDES

Capstone Project and Research Adviser


DANIEL MALIGAT JR., DIT

Capstone Project and Research Adviser

Approved:


MARJORIE S. LUCAS, MSc

Instructor I

College of Fisheries, Aquatic Sciences, and Technology

Appendix H

Letter for Final Defense

November 20, 2025

DANIEL E. MALIGAT JR., DIT
Capstone Project and Research 2 Professor
College of Computing and Multimedia Studies

Sir:

In partial fulfillment of the requirement in Bachelor of Science in Information Technology program, the undersigned conducted a Capstone Project entitled "PondMate: Fish Pond Management Mobile Application with IoT-Based Automatic Fish Feeder"

In connection with this, the undersigned humbly request the following CCMS faculty as members of our panel committee. Apart from that, we would like to schedule our final defense on December 10, 2025, at 3 AM/PM.

Thank you very much!

Panel Committee Members:

1. **NORIANNE C. LAMADRID**

Member

2. **EDGAR BRYAN E. NICART, DIT**

Member

Signature

Date

Signature

Date

Respectfully yours,

MACAPAGAL, RHYZELINE M.

VILLARANDA, VERGEL B.

LABRADOR, FRANCES FIONA P.

BABEJES, PAMELA JANE T.

Noted By:

SPENCER S. SALUDES

Capstone Project and Research Adviser

DANIEL MALIGAT JR., DIT

Capstone Project and Research Adviser

Appendix I

Sample of USE Questionnaire

QUESTIONNAIRE

PONDMATE: FISH POND MANAGEMENT MOBILE APPLICATION WITH IOT-BASED AUTOMATIC FISH FEEDER

Dear Respondents,

We are conducting a study to evaluate the effectiveness of a Fish Pond Management Mobile Application with an IoT-Based Automatic Fish Feeder. This questionnaire aims to gather your feedback on the system's usefulness, ease of use, ease of learning, and overall satisfaction. The information collected will be used solely for academic research purposes and will remain strictly confidential. Participation in this study is voluntary, and you may withdraw at any time without penalty. Please answer all questions honestly based on your experience. For statements that use a rating scale, kindly mark the box that best represents your opinion. Thank you for taking the time to participate — your feedback is valuable in improving fish pond management.

The Researchers

I. Fish Pond Profile

Instruction: Please check the answer based on your current situation.

1. How long have you been managing your fish pond?

☐ Less than 1 year
☐ 1–3 years
☐ 4–6 years
☐ More than 6 years

II. Pre-Deployment Evaluation

Instruction: Rate each statement according to your level of agreement.

	Scale
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

STATEMENT	5	4	3	2	1
I often forget the daily tasks needed for my fish pond management.					
Tracking fish growth, feeding, and pond conditions manually takes too much time.					
I sometimes overfeed or underfeed my fish due to lack of automation.					
I would benefit from a system that automatically schedules feeding and monitors pond activities.					
I believe technology can improve the efficiency of fish pond management.					
I am willing to adopt a mobile application and IoT-based automatic fish feeder.					

■. Post-Deployment Evaluation / Technology Acceptance Model (TAM)

Instruction: Rate your experience using the 1–5 scale above.

A. Perceived Usefulness (PU)

STATEMENT	5	4	3	2	1
The application helps me manage my fish pond more efficiently.					
The IoT feeder reduces errors in daily feeding.					
The system allows me to make better decisions in pond management.					

B. Perceived Ease of Use (PEOU)

STATEMENT	5	4	3	2	1
The application is easy to use.					
The IoT feeder is simple to operate and setup.					

C. Ease of Learning (EL)

STATEMENT	5	4	3	2	1
I quickly learned how to use the application.					
Instructions and guidance in the system are clear and helpful.					

D. Satisfaction (SAT)

STATEMENT	5	4	3	2	1
I am satisfied with the features of the application.					
I am satisfied with the IoT-based automatic feeder.					
Overall, I am satisfied using the system in my pond management.					

Appendix J

Arduino Mega Source Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <RTCLib.h>
#include "HX711.h"
#include <Servo.h>
#include <EEPROM.h>

// ===== LCD =====
LiquidCrystal_I2C lcd(0x27, 20, 4);

// ===== RTC =====
RTC_DS3231 rtc;

// ===== HX711 Pin Connections =====
#define DT 3    // Data pin (DT) of HX711 to pin 3 of Mega
#define SCK 2   // Clock pin (SCK) of HX711 to pin 2 of Mega
HX711 scale;

// ===== Calibration =====
float calibration_factor = 1150;    // Adjust this for
your load cell

const float weightThreshold = 5.0; // Minimum detectable
weight in grams

bool calibrationMode = false;       // Only true when 'w'
is pressed

bool scheduleReceived = false;

String lastSchedulePayload = "";

unsigned long lastScheduleRequest = 0;
```

```

const unsigned long scheduleRequestInterval = 30000; //
30 sec

// ===== SERVOS =====

Servo feederServoMG995; // MG995 (continuous rotation)
Servo feederServoSG90; // SG90 (standard servo)
#define SERVO_MG995_PIN 9
#define SERVO_SG90_PIN 8

void attachSG90() {
    feederServoSG90.attach(SERVO_SG90_PIN);
    delay(200);
}

void detachSG90() {
    feederServoSG90.detach();
}

// MG995 microsecond constants (tweak if needed)
const int MG_STOP_US    = 1500;
const int MG_FORWARD_US = 2000; // forward spin
const int MG_REVERSE_US = 1000; // reverse spin

// ===== L298N MOTOR DRIVER =====

#define ENB 10 // PWM speed control
#define IN3 11 // Direction
#define IN4 12 // Direction

// ===== Feeding Schedule =====

int feedingHours[] = {21, 12};

```

```

int feedingMinutes[] = {54, 0};
int feedingCount      = 2;

String pondName = "PONDIMATE"; // default if nothing
received

// To avoid retriggering
int lastFeedHour   = -1;
int lastFeedMinute = -1;

// ===== EEPROM ADDRESSES
// =====

#define ADDR_POND    0
#define ADDR_TIMES   50
#define ADDR_CALIB   200 // <-- new address for
calibration factor
#define ADDR_GRAMS   300 // new EEPROM address for grams

// ===== GLOBAL VAR FOR GRAMS
// =====

int targetGrams = 0; // grams to dispense (from ESP)

// when to start reversing (in grams before target)
float nearTargetThreshold = 24.0; // smaller, more
sensible default

// ----- Save string to EEPROM -----
void saveToEEPROM(int addr, String data) {
    for (int i = 0; i < data.length(); i++) {
        EEPROM.write(addr + i, data[i]);
    }

    EEPROM.write(addr + data.length(), '\0'); // null
terminator

```

```

}

// ----- Read string from EEPROM -----
String readFromEEPROM(int addr) {
    char data[100];
    int len = 0;
    unsigned char k;
    k = EEPROM.read(addr);
    while (k != '\0' && len < 99) {
        data[len++] = k;
        k = EEPROM.read(addr + len);
    }
    data[len] = '\0';
    return String(data);
}

// ----- Save float to EEPROM -----
void saveFloatToEEPROM(int addr, float value) {
    byte *data = (byte*)&value;
    for (int i = 0; i < sizeof(float); i++) {
        EEPROM.write(addr + i, data[i]);
    }
}

// ----- Read float from EEPROM -----
float readFloatFromEEPROM(int addr) {
    float value;
    byte *data = (byte*)&value;
    for (int i = 0; i < sizeof(float); i++) {
        data[i] = EEPROM.read(addr + i);
    }
}

```

```

        return value;
    }

// ----- Format time into hh:mm AM/PM -----
String formatTime(int hour, int minute) {
    String ampm;
    int displayHour;

    if (hour == 0) {
        displayHour = 12;
        ampm = "AM";
    } else if (hour < 12) {
        displayHour = hour;
        ampm = "AM";
    } else if (hour == 12) {
        displayHour = 12;
        ampm = "PM";
    } else {
        displayHour = hour - 12;
        ampm = "PM";
    }

    char buf[12];

    sprintf(buf, "%02d:%02d %s", displayHour, minute,
ampm.c_str());
    return String(buf);
}

// ----- Get next feeding time -----
String getNextFeeding(DateTime now) {
    for (int i = 0; i < feedingCount; i++) {

```



```

        if (now.hour() < feedingHours[i] ||
            (now.hour() == feedingHours[i] && now.minute() <
feedingMinutes[i])) {
            return formatTime(feedingHours[i],
feedingMinutes[i]);
        }
    }
    return formatTime(feedingHours[0], feedingMinutes[0]);
}

// ----- Helper: Parse "hh:mm am/pm" into hour/minute
-----

void parseTime(String t, int &hour, int &minute) {
    t.trim();
    t.toUpperCase();

    int colon = t.indexOf(':');
    if (colon < 0) return;

    hour = t.substring(0, colon).toInt();
    int minEnd = t.indexOf(' ', colon);
    if (minEnd < 0) minEnd = t.length();
    minute = t.substring(colon + 1, minEnd).toInt();

    if (t.endsWith("PM") && hour < 12) hour += 12;
    if (t.endsWith("AM") && hour == 12) hour = 0;
}

// ===== MOTOR CONTROL
=====

void motorForward(int speed) {
    digitalWrite(IN3, LOW);

```

```

    digitalWrite(IN4, HIGH);
    analogWrite(ENB, speed);
}

void motorStop() {
    analogWrite(ENB, 0);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
}

// ===== MOTOR FEEDING FUNCTION
=====

void runMotorFeeding() {
    analogWrite(ENB, 255);    // Max speed
    digitalWrite(IN3, LOW);  // Forward direction
    digitalWrite(IN4, HIGH);
    delay(60000);            // 1 minute
    motorStop();
}

// ===== START FEEDING
=====

void doFeeding() {
    Serial.print("Feeding started... Target: ");
    Serial.print(targetGrams);
    Serial.println(" g");

    // ==== MG995 Start Feeding Movement (Dispense Feed)
    ====

    unsigned long startTime = millis();

```

```

    // Start spinning forward (use microseconds for
consistent behavior)

    feederServoMG995.writeMicroseconds(MG_FORWARD_US);

    Serial.println("MG995: spinning forward...");

    while (true) {
        float currentWeight = scale.get_units();

        if (abs(currentWeight) < weightThreshold)
currentWeight = 0;

        Serial.print("Current weight: ");

        Serial.println(currentWeight);

        if (targetGrams > 0 && currentWeight >= (targetGrams -
nearTargetThreshold)) {

            Serial.println("⚠ Threshold reached → executing full
feeding sequence");

            // === Reverse MG995 first ===

            feederServoMG995.writeMicroseconds(MG_REVERSE_US);

            delay(1000);

            feederServoMG995.writeMicroseconds(MG_STOP_US);

            // === Start DC motor AFTER MG995 reverse ===

            digitalWrite(IN3, LOW);
            digitalWrite(IN4, HIGH);
            analogWrite(ENB, 255);    // <-- use valid PWM

            break;
        }

        // ===== Failsafe: 30 seconds max =====

```

```

        if (millis() - startTime > 30000UL) {
            Serial.println("⚠ Failsafe triggered! Stopping
MG995");
            feederServoMG995.writeMicroseconds(MG_STOP_US);
            break;
        }

        delay(50); // small delay for load cell stability

        // Failsafe: 30 seconds max
        if (millis() - startTime > 30000UL) {
            Serial.println("⚠ Failsafe triggered! Stopping
MG995");
            feederServoMG995.writeMicroseconds(MG_STOP_US); //
stop if failsafe
            break;
        }

        delay(50); // small delay for load cell stability
    }

    // ==== SG90 process ====
    Serial.println("SG90 opening...");
    attachSG90();

    feederServoSG90.write(160); // open (if your SG90 uses
this value; adjust if needed)

    delay(1000); // give enough time to
fully open

    detachSG90();

    // ==== RUN MOTOR FOR 1 MINUTE ====

```

```

runMotorFeeding();

Serial.println("SG90 closing...");
attachSG90();
feederServoSG90.write(0); // close (adjust if needed)
delay(1000);              // give enough time to fully
close
detachSG90();

Serial.println("✅ SG90 Closed. Feeding complete.");
}

void requestScheduleFromESP() {
    Serial1.println("REQ:SCHEDULE");
    Serial.println("Requesting schedule from ESP...");
}

// Robust RTC setter - call only when user issues command
void setRTCTime(int yr, int mon, int day, int hr, int
min, int sec) {
    // Basic sanity checks
    if (yr < 2000 || mon < 1 || mon > 12 || day < 1 || day
> 31 ||
        hr < 0 || hr > 23 || min < 0 || min > 59 || sec < 0
|| sec > 59) {
        Serial.println("❌ Invalid date/time values. Not
setting RTC.");
        return;
    }

    DateTime newt(yr, mon, day, hr, min, sec);
    rtc.adjust(newt);
}

```

```

Serial.print("⌂ RTC set to: ");
Serial.print(yr); Serial.print("/");
Serial.print(mon); Serial.print("/");
Serial.print(day); Serial.print(" ");
Serial.print(hr); Serial.print(":");
Serial.print(min); Serial.print(":");
Serial.println(sec);
}

// ===== SETUP =====
void setup() {
  Serial.begin(9600);    // Debug to PC
  Serial1.begin(9600);   // Communication with ESP8266

  lcd.init();
  lcd.backlight();

  if (!rtc.begin()) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("RTC not found!");
    while (1);
  }

  // HX711 setup with calibration
  scale.begin(DT, SCK);

  // Load calibration factor from EEPROM (if valid)
  float savedCalib = readFloatFromEEPROM(ADDR_CALIB);
  if (!isnan(savedCalib) && savedCalib > 0.0) {
    calibration_factor = savedCalib;
  }
}

```

```

    Serial.print("Loaded calibration factor from EEPROM:
");
    Serial.println(calibration_factor);
} else {
    Serial.print("Using default calibration factor: ");
    Serial.println(calibration_factor);
}

scale.set_scale(calibration_factor);
scale.tare(); // Reset to 0

Serial.println("HX711 Load Cell Ready...");

Serial.println("Press '+' or '-' in Serial Monitor to
adjust calibration.");
Serial.println("Press 'w' to START calibration mode,
's' to STOP.");

// Attach MG995 and set neutral
feederServoMG995.attach(SERVO_MG995_PIN);
feederServoMG995.writeMicroseconds(MG_STOP_US);

// Ensure SG90 is closed at startup
attachSG90();
feederServoSG90.write(0); // close (adjust if needed)
delay(500);
detachSG90();

// Motor setup
pinMode(ENB, OUTPUT);
pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);

```

```

motorStop();

// === Load pond name + times from EEPROM ===
String savedPond = readFromEEPROM(ADDR_POND);
String savedTimes = readFromEEPROM(ADDR_TIMES);

int savedGrams;                                // <-- declare
here

EEPROM.get(ADDR_GRAMS, savedGrams);            // <-- read
grams from EEPROM

if (savedGrams > 0) {
    targetGrams = savedGrams;                    // <-- restore
global variable

    Serial.print("Loaded grams: ");
    Serial.println(targetGrams);
}

if (savedPond.length() > 0) pondName = savedPond;
if (savedTimes.length() > 0) {
    int firstComma = savedTimes.indexOf(',');
    String t0 = savedTimes.substring(0, firstComma);
    String t1 = savedTimes.substring(firstComma + 1);

    parseTime(t0, feedingHours[0], feedingMinutes[0]);
    parseTime(t1, feedingHours[1], feedingMinutes[1]);
}

lcd.setCursor(0, 0);
lcd.print("Pond Name: ");

lcd.print(pondName.substring(0, 20 - 11)); // keep
within 20 chars

```



```

String feedingLabel = "Feeding Schedule:";
lcd.setCursor((20 - feedingLabel.length()) / 2, 2);
lcd.print(feedingLabel);

}

void handleManualGrams(String cmd) {
    if (cmd.startsWith("grams ")) {
        String val = cmd.substring(6);
        val.trim();
        int g = val.toInt();

        if (g > 0) {
            targetGrams = g;
            EEPROM.put(ADDR_GRAMS, targetGrams);

            Serial.print("✚ Manual grams set to: ");
            Serial.print(targetGrams);
            Serial.println(" g (saved to EEPROM)");
        }
        else {
            Serial.println("✕ Invalid grams. Example: grams
25");
        }
    }
}

// ===== LOOP =====
void loop() {
    DateTime now = rtc.now();

```

```

// ===== AUTO REQUEST SCHEDULE =====
if (!scheduleReceived && millis() - lastScheduleRequest
>= scheduleRequestInterval) {
    requestScheduleFromESP();
    lastScheduleRequest = millis();
}

// ===== HX711 Reading =====
scale.set_scale(calibration_factor);
float currentWeight = scale.get_units();

if (abs(currentWeight) < weightThreshold) {
    currentWeight = 0;
}

// ===== SERIAL INPUT HANDLING (WITH COMMAND SYSTEM)
=====
if (Serial.available()) {
    String cmd = Serial.readStringUntil('\n');
    cmd.trim();
    cmd.toLowerCase();

// ===== CALIBRATION CONTROLS =====
if (cmd == "+") {
    calibration_factor += 100;
    saveFloatToEEPROM(ADDR_CALIB, calibration_factor);
    Serial.print("Calibration factor increased: ");
    Serial.println(calibration_factor);
}
else if (cmd == "-") {
    calibration_factor -= 100;

```

```

        saveFloatToEEPROM(ADDR_CALIB, calibration_factor);
        Serial.print("Calibration factor decreased: ");
        Serial.println(calibration_factor);
    }
    else if (cmd == "w") {
        calibrationMode = true;
        Serial.println("Calibration mode STARTED...");
    }
    else if (cmd == "s") {
        calibrationMode = false;
        Serial.println("Calibration mode STOPPED.");
    }

    // ===== NEW: MANUAL FEED COMMAND =====
    else if (cmd == "feed") {
        Serial.println("🤖 Manual feed command received!");
        Serial.print("Using saved target grams: ");
        Serial.println(targetGrams);

        doFeeding();    // <-- run the full feeding sequence
    }

    // ===== MANUAL RTC TIME COMMAND =====
    // Command format: settime YYYY/MM/DD HH:MM:SS
    else if (cmd.startsWith("settime")) {
        String param = cmd.substring(7); // remove "settime"
        param.trim();
        int spaceIndex = param.indexOf(' ');
        if (spaceIndex <= 0) {
            Serial.println("❌ Use: settime 2025/12/10
01:45:00");
        } else {

```

```

String datePart = param.substring(0, spaceIndex);
String timePart = param.substring(spaceIndex + 1);

// Parse date robustly (accepts YYYY/MM/DD or YYYY-
MM-DD)

datePart.replace('-', '/');
int d1 = datePart.indexOf('/');
int d2 = datePart.indexOf('/', d1 + 1);
if (d1 < 0 || d2 < 0) {
    Serial.println("✗ Date format wrong. Use
YYYY/MM/DD");
} else {
    int yr = datePart.substring(0, d1).toInt();
    int mon = datePart.substring(d1 + 1, d2).toInt();
    int day = datePart.substring(d2 + 1).toInt();

    // Parse time robustly using colon positions
    (accepts 1 or 2 digit hour)

    int c1 = timePart.indexOf(':');
    int c2 = timePart.lastIndexOf(':');
    if (c1 < 0 || c2 == c1) {
        Serial.println("✗ Time format wrong. Use
HH:MM:SS");
    } else {
        int hr = timePart.substring(0, c1).toInt();
        int mn = timePart.substring(c1 + 1, c2).toInt();
        int sc = timePart.substring(c2 + 1).toInt();

        setRTCTime(yr, mon, day, hr, mn, sc);
    }
}
}

```

```

}

// ===== NEW COMMAND: DC MOTOR TEST =====
else if (cmd == "dc") {
    Serial.println("⚙️ DC Motor Test: Spinning for 2
seconds...");

    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
    analogWrite(ENB, 255);

    delay(50000);    // 2 seconds spin

    motorStop();

    Serial.println("🛑 DC Motor Stopped.");
}
else if (cmd == "time") {
    DateTime now = rtc.now();
    Serial.print("RTC: ");
    Serial.print(now.year()); Serial.print("/");
    Serial.print(now.month()); Serial.print("/");
    Serial.print(now.day()); Serial.print(" ");
    Serial.print(now.hour()); Serial.print(":");
    if (now.minute() < 10) Serial.print('0');
    Serial.print(now.minute()); Serial.print(":");
    if (now.second() < 10) Serial.print('0');
    Serial.println(now.second());
}

```

```

// ===== EXTRA SAMPLE COMMANDS (OPTIONAL) =====

else if (cmd == "mgf") {
    Serial.println("MG995: Forward spin 1 sec");
    feederServoMG995.writeMicroseconds(MG_FORWARD_US);
    delay(1000);
    feederServoMG995.writeMicroseconds(MG_STOP_US);
}

else if (cmd == "mgr") {
    Serial.println("MG995: Backward spin 1 sec");
    feederServoMG995.writeMicroseconds(MG_REVERSE_US);
    delay(1000);
    feederServoMG995.writeMicroseconds(MG_STOP_US);
}

else if (cmd == "open") {
    Serial.println("SG90: Opening...");
    attachSG90();
    feederServoSG90.write(160);
    delay(700);
    detachSG90();
}

else if (cmd == "close") {
    Serial.println("SG90: Closing...");
    attachSG90();
    feederServoSG90.write(0);
    delay(700);
    detachSG90();
}

```

```

}

else if (cmd == "stop") {

    Serial.println("🛑 STOP COMMAND RECEIVED!");

    feederServoMG995.writeMicroseconds(MG_REVERSE_US);
    delay(1000);
    feederServoMG995.writeMicroseconds(MG_STOP_US);

    Serial.println("SG90: Closing...");
    attachSG90();
    feederServoSG90.write(0);
    delay(700);
    detachSG90();

    motorStop();

    Serial.println("🛑 SYSTEM STOPPED.");
}

else {

    // Check if it is a grams command
    handleManualGrams(cmd);
}

}

// ==== ONLY PRINT WHEN IN CALIBRATION MODE ====
if (calibrationMode) {

```

```

    Serial.print("Weight: ");
    Serial.print((int)currentWeight);    // now integer
only
    Serial.print(" g | Calibration Factor: ");
    Serial.println(calibration_factor);
}

// ==== LISTEN TO ESP MESSAGES (Serial1) ====
if (Serial1.available()) {
    String line = Serial1.readStringUntil('\n');
    line.trim();

    if (line.startsWith("SEND:")) {
        // ☒ Stop auto requesting (we received a schedule)
        scheduleReceived = true;

        // ☒ Check if same or new payload
        if (line == lastSchedulePayload) {
            Serial.println("▶▶ Same schedule received.
Ignoring...");
            return; // do nothing
        }

        Serial.println("☒ New schedule received from
ESP!");
        Serial.println(line);

        lastSchedulePayload = line; // Save new schedule to
compare next time

        line.remove(0, 5);

```



```

int sep1 = line.indexOf('|');
int sep2 = line.indexOf('|', sep1 + 1);

if (sep1 > 0 && sep2 > sep1) {
    pondName = line.substring(0, sep1);
    String times = line.substring(sep1 + 1, sep2);
    String gramsStr = line.substring(sep2 + 1);

    // ☒ update global grams
    targetGrams = gramsStr.toInt();
    EEPROM.put(ADDR_GRAMS, targetGrams);    // <--
    save grams persistently

    int firstComma = times.indexOf(',');

    String t0 = times.substring(0, firstComma);
    String t1 = times.substring(firstComma + 1);

    parseTime(t0, feedingHours[0],
feedingMinutes[0]);
    parseTime(t1, feedingHours[1],
feedingMinutes[1]);

    saveToEEPROM(ADDR_POND, pondName);
    saveToEEPROM(ADDR_TIMES, times);

    lcd.setCursor(0, 0);
    lcd.print("                ");
    lcd.setCursor(0, 0);
    lcd.print("Pond Name: ");
    lcd.print(pondName.substring(0, 20));

```

```

        Serial.print("☑ Schedule updated & saved! Grams:
");

        Serial.println(targetGrams);
    }
}

// ==== TIME DISPLAY ====
static String lastTime = "";
String timeStr = "Time: " + formatTime(now.hour(),
now.minute());
if (timeStr != lastTime) {
    lcd.setCursor(0, 1);
    lcd.print("                ");
    lcd.setCursor((20 - timeStr.length()) / 2, 1);
    lcd.print(timeStr);
    lastTime = timeStr;
}

// ==== LINE 4: Cycle through all feeding times ====
static unsigned long lastSwitch = 0;
static int feedIndex = 0;

if (millis() - lastSwitch >= 5000) { // change every 5
seconds
    String feedStr = "T" + String(feedIndex+1) + ": " +
        formatTime(feedingHours[feedIndex],
feedingMinutes[feedIndex]);

    lcd.setCursor(0, 3);
    lcd.print("                ");
    lcd.setCursor((20 - feedStr.length()) / 2, 3);

```

```

    lcd.print(feedStr);

    feedIndex = (feedIndex + 1) % feedingCount;

    lastSwitch = millis();
}

// SG90 AUTO-RESET: 1 SEC BEFORE FEEDING
for (int i = 0; i < feedingCount; i++) {
    if (now.hour() == feedingHours[i] && now.minute() ==
feedingMinutes[i] && now.second() == 59) {

        Serial.println("🔄 Auto-reset SG90 before
feeding...");

        attachSG90();

        feederServoSG90.write(0); // Close SG90

        delay(1000); // Hold closed for 1
second

        detachSG90();
    }
}

// ==== Feeding check ====
for (int i = 0; i < feedingCount; i++) {
    if (now.hour() == feedingHours[i] && now.minute() ==
feedingMinutes[i]) {

        if (!(lastFeedHour == now.hour() && lastFeedMinute
== now.minute())) {

            doFeeding();

            lastFeedHour = now.hour();

            lastFeedMinute = now.minute();

        }
    }
}

delay(300);
}

```

Appendix K

ESP 8266 Source Code

```
#include <ESP8266WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
#include <SoftwareSerial.h>

// ==== Debug Serial (pins D1=GPIO5 RX, D2=GPIO4 TX) ====
SoftwareSerial Debug(5, 4);

// ==== WiFi and Adafruit IO Credentials ====
const char* WIFI_SSID = "GlobeAtHome_4B4AC";
const char* WIFI_PASS = "pondmate";
#define AIO_SERVER      "io.adafruit.com"
#define AIO_SERVERPORT  1883
#define AIO_USERNAME    "ver_lev"
#define AIO_KEY
"aio_QyAU79SOyUkiRw4LJly79rdLEV0Z"

WiFiClient client;

Adafruit_MQTT_Client mqtt(&client, AIO_SERVER,
AIO_SERVERPORT, AIO_USERNAME, AIO_KEY);

// Subscribe to schedule feed

Adafruit_MQTT_Subscribe feedTimes =
Adafruit_MQTT_Subscribe(&mqtt, AIO_USERNAME
"/feeds/schedule");

String lastPayload = "";

void printWiFiStatus() {
  if (WiFi.status() == WL_CONNECTED) {
    Debug.print("✅ WiFi Connected | IP: ");
    Debug.println(WiFi.localIP());
  } else {
    Debug.println("❌ WiFi NOT Connected!");
  }
}
```

```

    }
}

void setup() {
    Debug.begin(9600);
    Serial.begin(9600);
    Debug.println("Connecting to WiFi...");
    WiFi.begin(WIFI_SSID, WIFI_PASS);
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Debug.print(".");
    }

    Debug.println("\n✅ WiFi Connected!");
    Debug.print("IP: ");
    Debug.println(WiFi.localIP());
    mqtt.subscribe(&feedTimes);
}

void loop() {
    // ===== AUTO RESTART ESP IF WIFI LOST =====
    static unsigned long lastWiFiCheck = 0;
    const unsigned long wifiCheckInterval = 10000; // every
10s
    static unsigned long wifiDisconnectedSince = 0;
    static bool wasConnected = true;
    if (millis() - lastWiFiCheck > wifiCheckInterval) {
        lastWiFiCheck = millis();
        printWiFiStatus();
        if (WiFi.status() != WL_CONNECTED) {
            if (wasConnected) {
                wifiDisconnectedSince = millis();
                wasConnected = false;
            }
        }
    }
}

```

```

    }

    Debug.println("⚠️ WiFi Lost... reconnecting...");
    WiFi.reconnect();

    // Restart ESP if not reconnected for 20s
    if (millis() - wifiDisconnectedSince > 20000) {
        Debug.println("🔴 No WiFi for 30s → Restarting ESP!");
        ESP.restart();
    }
} else {
    wasConnected = true;
}
}

// ===== MQTT CONNECTION =====
if (!mqtt.connected()) {
    connectToMQTT();
}

mqtt.processPackets(10);
mqtt.ping();

// ===== LISTEN FOR REQUEST FROM MEGA =====
if (Serial.available()) {
    String req = Serial.readStringUntil('\n');
    req.trim();

    if (req == "REQ:SCHEDULE") {
        Debug.println("📡 Mega requested schedule...");

        if (lastPayload != "") {
            Serial.println(lastPayload);
        }
    }
}

```

```

        Debug.println("📧 Resent: " + lastPayload);
    } else {
        Debug.println("⚠️ No schedule to resend yet");
    }
}

// ===== LISTEN FOR NEW SCHEDULE FROM ADAFRUIT =====
Adafruit_MQTT_Subscribe *subscription;
while ((subscription = mqtt.readSubscription(10))) {
    if (subscription == &feedTimes) {
        char *msg = (char *)feedTimes.lastread;
        String data = String(msg);

        int sep1 = data.indexOf('|');
        int sep2 = data.indexOf('|', sep1 + 1);

        if (sep1 > 0 && sep2 > sep1) {
            String pondName = data.substring(0, sep1);
            String times = data.substring(sep1 + 1, sep2);
            String grams = data.substring(sep2 + 1);
            grams.trim();

            Debug.println("📧 New schedule received!");
            Debug.println("Pond: " + pondName);
            Debug.println("Times: " + times);
            Debug.println("Grams: " + grams);

            // 📧 FORMAT & SEND TO MEGA

```

```

        String payload = "SEND:" + pondName + "|" + times
+ "|" + grams;

        if (payload != lastPayload) {
            Serial.println(payload);

            Debug.println("📡 Sent to Mega: " + payload);
            lastPayload = payload;
        } else {
            Debug.println("⏮ Same schedule, not sending
again");
        }

    } else {
        Debug.println("📡 Raw schedule: " + data);
    }
}

}

void connectToMQTT() {
    Debug.print("Connecting to Adafruit IO...");
    int8_t ret;
    while ((ret = mqtt.connect()) != 0) {
        Debug.println(mqtt.connectErrorString(ret));
        Debug.println("Retrying in 5 seconds...");
        mqtt.disconnect();
        delay(5000);
    }

    Debug.println("✅ Connected to Adafruit IO!");
}

```


CURRICULUM VITAE

Rhyzeline M. Macapagal

Purok 4, Brgy. 2
Mercedes, Camarines Norte
09488380580
rhyzelinemacapagal@gmail.com



SUMMARY:

An IT undergraduate from Camarines Norte State College with a solid background in Java, Python, Visual Basic, HTML, CSS, JavaScript, PHP, and MySQL database management. Experienced in utilizing Microsoft Office Suite for documentation, data processing, and creating professional presentations. Proficient in Adobe Creative Suite for video editing, animation, and other multimedia tasks. Skilled in 3D modeling and visualization using 3ds Max and SketchUp.

PERSONAL BACKGROUND:

DATE OF BIRTH:	October 21, 2004
PLACE OF BIRTH:	Daet, Camarines Norte
AGE:	21
SEX:	Female
HEIGHT:	5'6"
WEIGHT:	49 kgs.
CIVIL STATUS:	Single
CITIZENSHIP:	Filipino
RELIGION:	Roman Catholic

EDUCATIONAL BACKGROUND:

TERTIARY:	Camarines Norte State College A.Y. 2022 - 2026 BS Information Technology Daet, Camarines Norte
SECONDARY:	Mabini Colleges, Inc.

Daet, Camarines Norte
Mercedes High School
A.Y. 2016 - 2020
Mercedes, Camarines Norte

PRIMARY: Mercedes Elementary School
A.Y. 2010 - 2016
Mercedes, Camarines Norte

TECHNICAL SKILL

PROGRAMMING

- Proficient in Java, Python, Visual Basic, HTML, CSS, JavaScript, PHP, and MySQL for building applications, websites, and database-driven systems.

ADOBE CREATIVE SUITE

- Skilled in using Adobe After Effects, Photoshop, and Premiere Pro for producing high-quality video edits and animation.

3D Modeling & Design

- Proficient in 3ds Max and SketchUp for creating detailed 3D models, visualizations, and design concepts.

PERSONAL CHARACTERISTICS

- Responsible and trustworthy
 - Motivated and hardworking
 - Positive attitude and strong work ethic
 - Creative and resourceful
-

SEMINARS AND CONFERENCE ATTENDED

- BITSCON Bicol IT Student Congress 2024 April 25 - 26, 2024
Camarines Norte State College, Daet, Camarines Norte
 - FROM SCROLL TO SKILL: Empowering Students and Job Seekers through AI
March 14, 2025
-

REFERENCES:

- **DANIEL E. MALIGAT JR., DIT**
CCMS, Dean

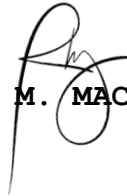
Camarines Norte State College
Daet, Camarines Norte

- **SPENCER S. SALUDES**

Instructor

Camarines Norte State College

I hereby certify that the pieces of information presented above are true and correct to the best of my knowledge and belief


RHYZELINE M. MACAPAGAL

Vergel B. Villaranda

Purok 2, Brgy. Hinipaan
Mercedes, Camarines Norte
09668223328
vergelvillaranda@gmail.com



SUMMARY:

An IT undergraduate from Camarines Norte State College capable in both software and hardware development, with experience using Java, Visual Basic, HTML, CSS, JavaScript, MySQL, and Firebase to build functional and responsive application and websites. I also work on Arduino projects that integrate sensors, modules, and real-time systems. Alongside programming, I am proficient in 3D modeling, sketching, and designing 3D simulations, allowing me to create detailed prototypes and visual representations for technical projects.

PERSONAL BACKGROUND:

DATE OF BIRTH:	September 9, 2004
PLACE OF BIRTH:	Mercedes, Camarines Norte
AGE:	23
SEX:	Male
HEIGHT:	5'10"
WEIGHT:	70 kgs.
CIVIL STATUS:	Single
CITIZENSHIP:	Filipino
RELIGION:	Roman Catholic

EDUCATIONAL BACKGROUND:

TERTIARY:	Camarines Norte State College A.Y. 2022 - 2026 BS Information Technology Daet, Camarines Norte
SECONDARY:	Mabini Colleges, Inc. S.Y. 2018 - 2020

Pablo S. Villafuerte High
School
A.Y. 2014 - 2018
Mercedes, Camarines Norte

PRIMARY:

Hinipaan Elementary School
A.Y. 2009 - 2014
Mercedes, Camarines Norte

TECHNICAL SKILL

PROGRAMMING

- Knowledgeable in using languages such as Java, Visual Basic, HTML, CSS, JavaScript, MySQL, and Firebase to build applications and solve problems.

EMBEDDED SYSTEMS DEVELOPMENT

- Hands-on experience working with Arduino projects, integrating sensors, modules, and hardware components.

ADOBE CREATIVE SUITE

- Capable of editing photos and videos using Adobe tools to create clean, high-quality visuals and transitions.

3D MODELING AND DESIGN

- Familiar with creating 3D models, sketches, and simulations for prototypes and project visualization

PERSONAL CHARACTERISTICS

- Organized and Goal-driven
 - Eager to learn and a problem solver
 - Responsible and Hardworking
 - Reliable and Flexible
-

SEMINARS AND CONFERENCE ATTENDED

- BITSCON Bicol IT Student Congress 2024 April 25 - 26, 2024
Camarines Norte State College, Daet, Camarines Norte
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March 14, 2025

REFERENCES:

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CCMS, Dean
Camarines Norte State College Daet, Camarines Norte
- **SPENCER S. SALUDES**
Instructor
Camarines Norte State College

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VERGEL B. VILLARANDA

Frances Fiona P. Labrador

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Daet, Camarines Norte
09705289853
francesfionalabrador@gmail.com



SUMMARY:

An IT undergraduate from Camarines Norte State College with expertise in Java, Visual Basic, basic Python, PHP, MySQL, 3D graphic design using 3Ds Max, and game development. Skilled in UI design using Figma, poster creation, and video editing with Adobe Creative Suite. A quick learner and analytical thinker with the ability to work effectively both independently and within team settings.

PERSONAL BACKGROUND:

DATE OF BIRTH:	September 20, 2003
PLACE OF BIRTH:	Daet, Camarines Norte
AGE:	22
SEX:	Female
HEIGHT:	5'4"
WEIGHT:	45 kgs.
CIVIL STATUS:	Single
CITIZENSHIP:	Filipino
RELIGION:	Roman Catholic

EDUCATIONAL BACKGROUND:

TERTIARY:	Camarines Norte State College A.Y. 2022 – 2026 BS Information Technology Daet, Camarines Norte
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SECONDARY:	Vinzons Pilot High School S.Y. 2020 – 2022 Senior High School
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Vinzons Pilot High School
A.Y. 2016 - 2020
Vinzons, Camarines Norte

PRIMARY: Daet Elementary School
A.Y. 2010 - 2016
Daet, Camarines Norte

TECHNICAL SKILL

PROGRAMMING

- Knowledgeable in Java, Visual Basic, basic Python, PHP, and MySQL for application development and database management.

UI/UX DESIGN

- Experienced in Figma for creating user interfaces and designing visually appealing layouts.

ADOBE CREATIVE SUITE

- Skilled in Adobe Photoshop and Premiere Pro for poster creation, photo editing, and video production.

3D MODELING & DESIGN

- Proficient in 3ds Max for creating 3D models, visualizations, and design concepts.

PERSONAL CHARACTERISTICS

-
- Quick learner with strong analytical and problem-solving skills
 - Adaptable and able to work effectively both independently and in teams
 - Detail-oriented with a creative mindset
 - Strong communication and collaboration abilities
 - Highly motivated and committed to continuous learning
-

SEMINARS AND CONFERENCE ATTENDED

- BITSCON Bicol IT Student Congress 2024 April 25 - 26, 2024
Camarines Norte State College, Daet, Camarines Norte
 - FROM SCROLL TO SKILL: Empowering Students and Job Seekers through AI
March 14, 2025
-

REFERENCES:

- **DANIEL E. MALIGAT JR., DIT**
CCMS, Dean
Camarines Norte State College
Daet, Camarines Norte
- **SPENCER S. SALUDES**
Instructor
Camarines Norte State College

I hereby certify that the pieces of information presented above are true and correct to the best of my knowledge and belief.


FRANCES FIONA P. LABRADOR

Pamela Jane T. Babejes

Blk 16 Lot 4 Rosal Street, Fairview 3
Subdivision, Brgy. Magang, Daet,
Camarines Norte
09957863021
pamelajanetopularbabejes@gmail.com



SUMMARY:

An IT undergraduate from Camarines Norte State College motivated and adaptable with strong skills in documentation, basic front-end development, and foundational 3D modelling. Able to interpret code logic and translate system requirement into clear visual and structural representations. Capable of organizing technical information and supporting both technical and creative aspects of development. Seeking opportunities to apply and further strengthen skills in software development, multimedia work, or system administration.

PERSONAL BACKGROUND:

DATE OF BIRTH:	January 24, 2004
PLACE OF BIRTH:	Daet, Camarines Norte
AGE:	21
SEX:	Female
HEIGHT:	5'3"
WEIGHT:	69 kgs.
CIVIL STATUS:	Single
CITIZENSHIP:	Filipino
RELIGION:	Born-Again Christian

EDUCATIONAL BACKGROUND:

TERTIARY:	Camarines Norte State College A.Y. 2022 - 2026 BS Information Technology Daet, Camarines Norte
------------------	---

SECONDARY:	Camarines Norte Senior High School S.Y. 2020 - 2022 Senior High School
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Camarines Norte National
High School
A.Y. 2016 - 2020
Vinzons, Camarines Norte

PRIMARY:

Daet Elementary School
A.Y. 2015 - 2016
Daet, Camarines Norte

Pambuhan Elementary School
A.Y. 2010 - 2015
Pambuhan, Mercedes, Camarines Norte

TECHNICAL SKILL

PROGRAMMING

- Understanding of code logic with experience in Java, basic Python, and basic database handling such as PHP and MYSQL.

UI/UX DESIGN

- Basic knowledge of layout planning, user flow, and simple interface creation.

DOCUMENTATION

- Proficient in Microsoft Word, academic writing, structured research papers/thesis paper, and technical documentation.

VISUAL SYSTEM DESIGN

- Able to create clear visual representations of system processes and structures, including organized diagrams and workflow visuals.

3D MODELING & DESIGN

- Basic experience in creating and modifying 3D objects and simple environments using Autodesk 3Ds Max.

PERSONAL CHARACTERISTICS

- Strong attention to detail, especially in documentation and visual design
- Organized and able to plan structured workflows and technical tasks
- Good analytical thinking and capable of understanding complex system logic
- Effective communicator, both in writing and teamwork settings

- Creative mindset with the ability to combine technical and design-oriented work

SEMINARS AND CONFERENCE ATTENDED

- BITSCON Bicol IT Student Congress 2024 April 25 - 26, 2024
Camarines Norte State College, Daet, Camarines Norte
- FROM SCROLL TO SKILL: Empowering Students and Job Seekers through AI
March 14, 2025
- Bridging the Digital and Physical: 3D Tech for Visionaries
May 7, 2025

REFERENCES:

- **DANIEL E. MALIGAT JR., DIT**
CCMS, Dean
Camarines Norte State College Daet, Camarines Norte
- **SPENCER S. SALUDES**
Instructor
Camarines Norte State College

I hereby certify that the pieces of information presented above are true and correct to the best of my knowledge and belief.



PAMELA JANE T. BABEJES